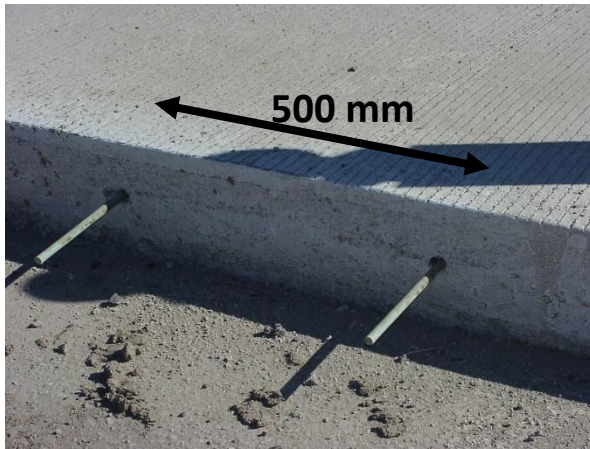


ACI-239C Internal report  
Flexural Testing and Back-calculation of Tensile  
properties

Barzin Mobasher  
Arizona State University  
[barzin@asu.edu](mailto:barzin@asu.edu)

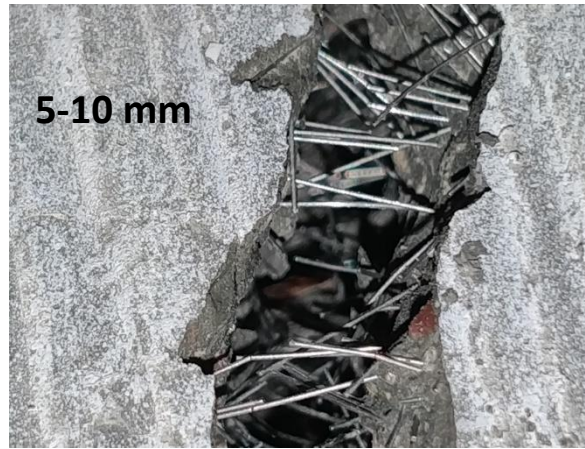
# Why Should we move towards Hybrid reinforcement strategies?

Strengthening at the microstructure level will have a profound impact at the macro-structural level strength and durability



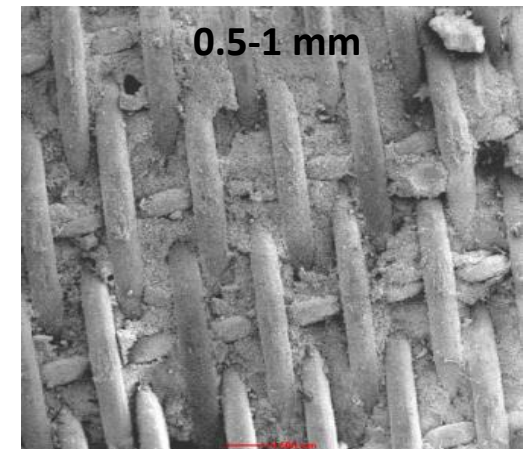
Ordinary reinforced concrete slab, 5-10" thick, standard rebars, 12"-18" apart

**Reinforcing Bars:**  
Structural reinforcement



Fiber reinforced Concrete (FRC), reinforced at the meso-structure with  $\frac{1}{2}$ " long fibers,  $\frac{1}{2}$ " (10mm) millimeters apart

**Fiber reinforced Concrete:**  
Meso-structure reinforcement



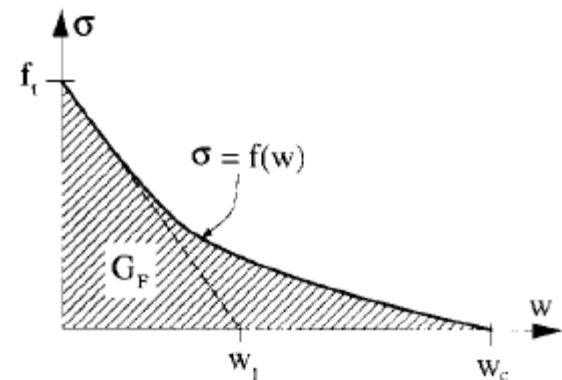
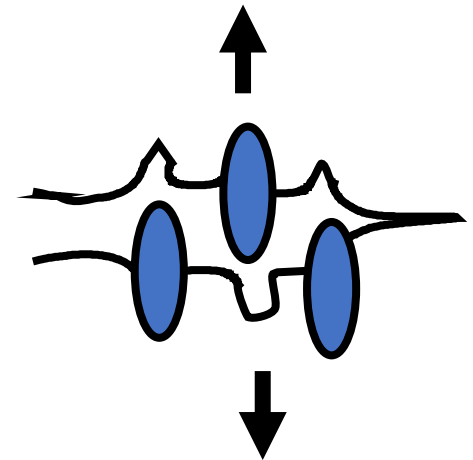
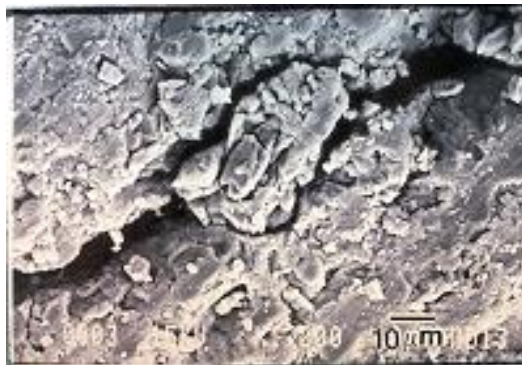
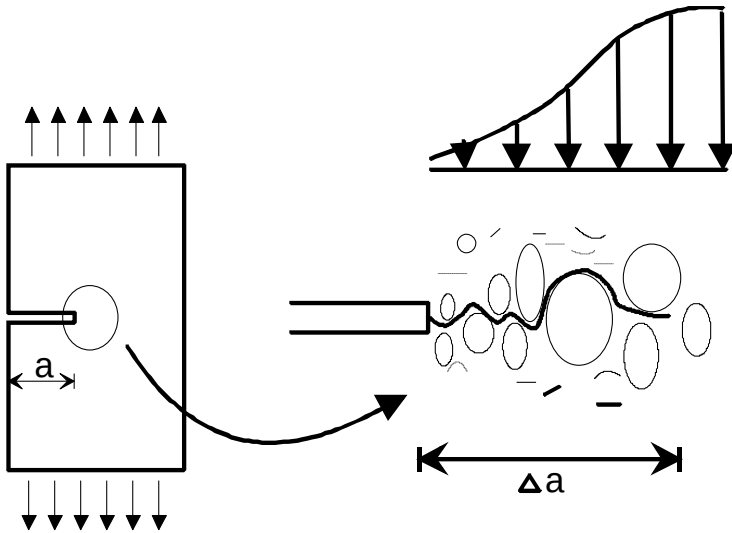
Textile reinforced Concrete (TRC), reinforced at the micro-structure with long fibers, woven, and only millimeters apart

**Textile Reinforced Concrete:**  
Micro-structure reinforcement

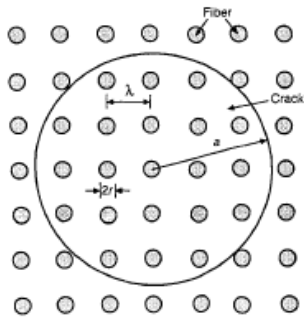
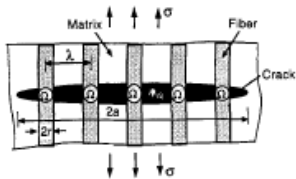
UHPC

# Fracture Process in Quasi-Brittle Materials

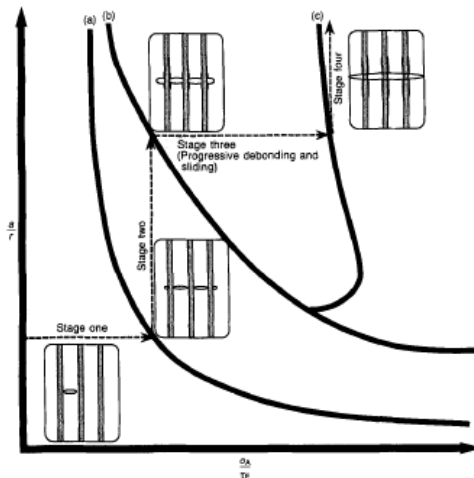
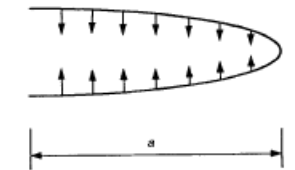
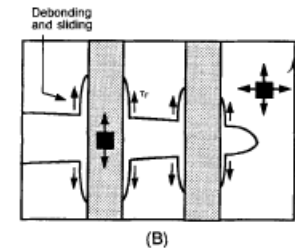
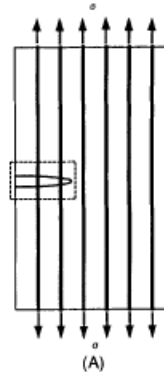
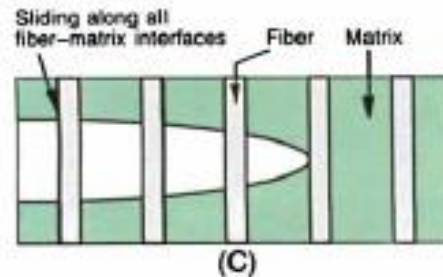
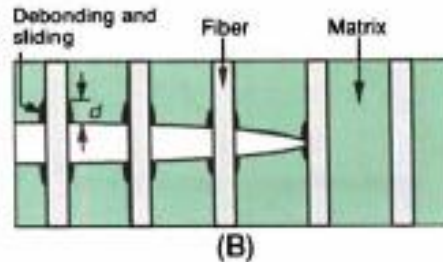
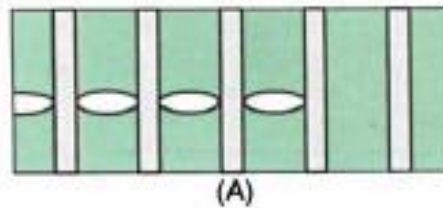
1. Stable Crack growth
2. Fracture Mechanics is applicable
3. Energy loss as the crack grows, dissipation along the crack path



# Micromechanics of toughening- why Fibers increase the tensile strength and ductility ?

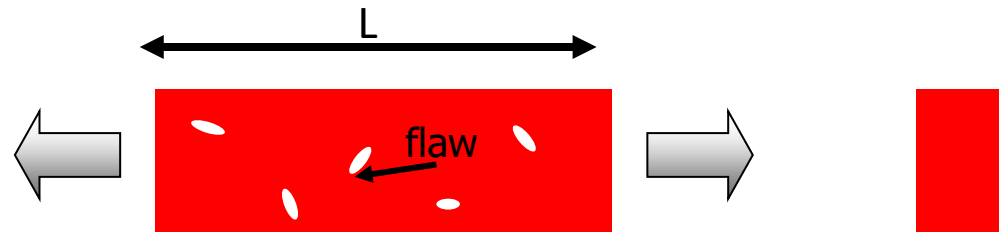


- Crack Bridging mechanisms
- Three parameters of “Stress”, “Crack opening”, “Crack length” are important tools of modeling,
- $\sigma$ - $\omega$ - $a$ , or  $\sigma(\omega)$ ,  $\sigma(a)$ ,  $\omega(a)$

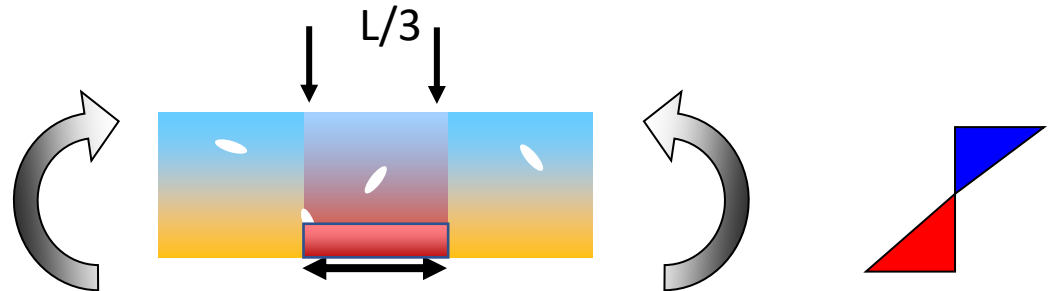


# Uniaxial Tension vs. Flexural Bending

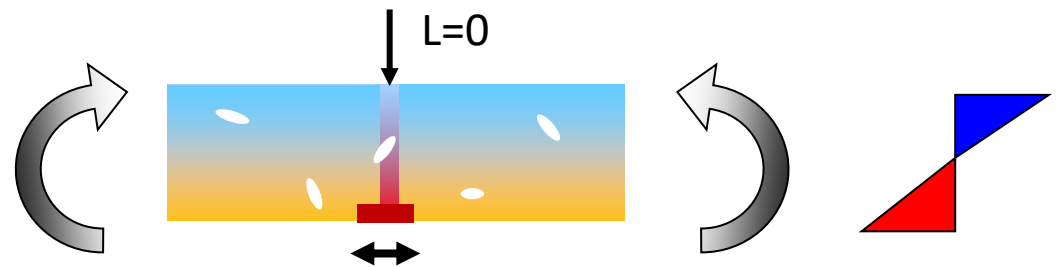
- Uniaxial Tension



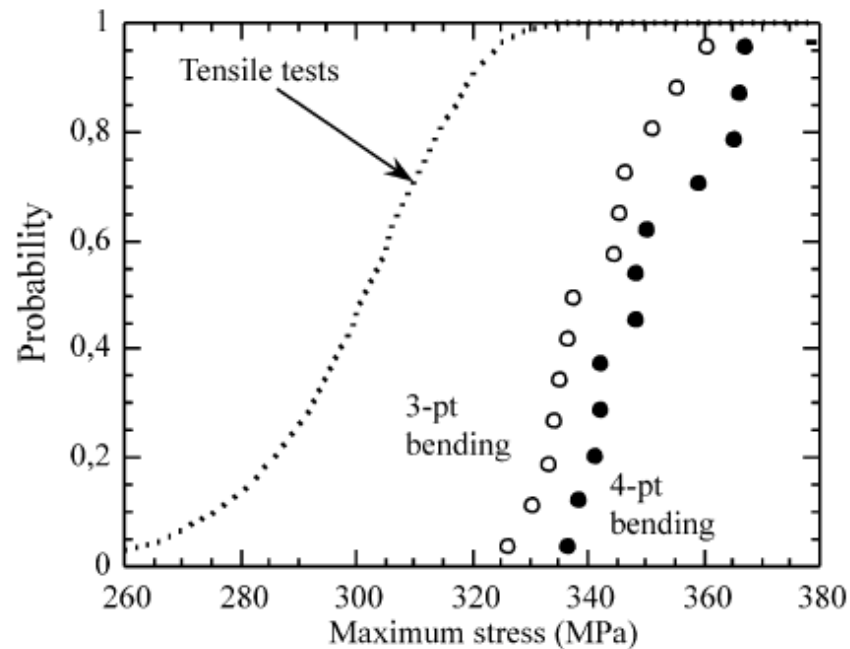
- Flexural Bending (4 point bending)



- Flexural Bending (3 point bending)



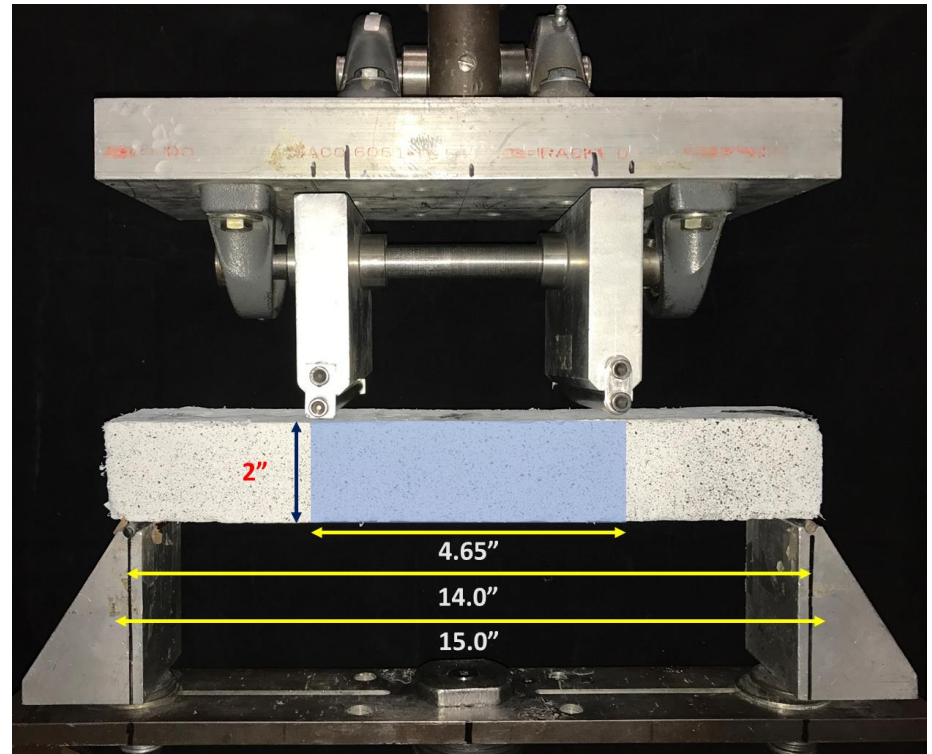
## Size Effect-Fundamental differences in tension notched or unnotched, and flexural test

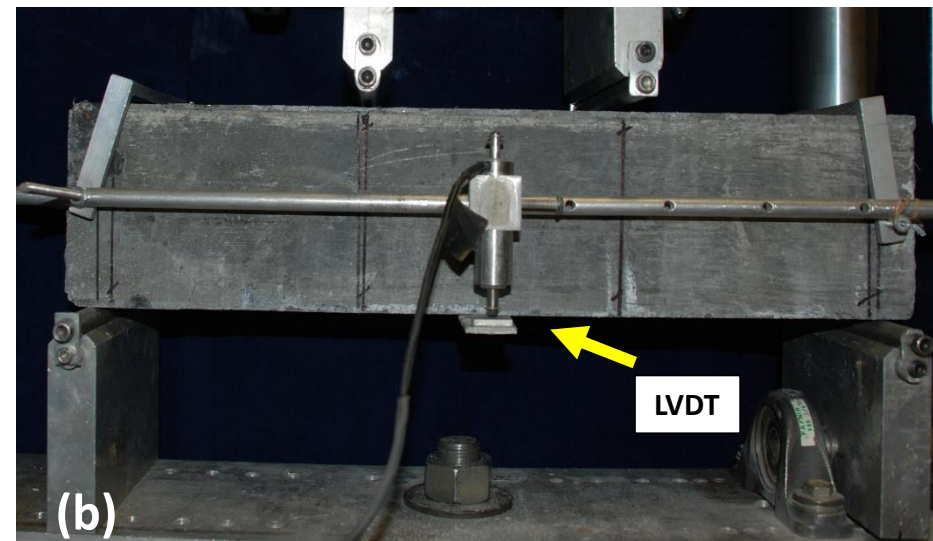
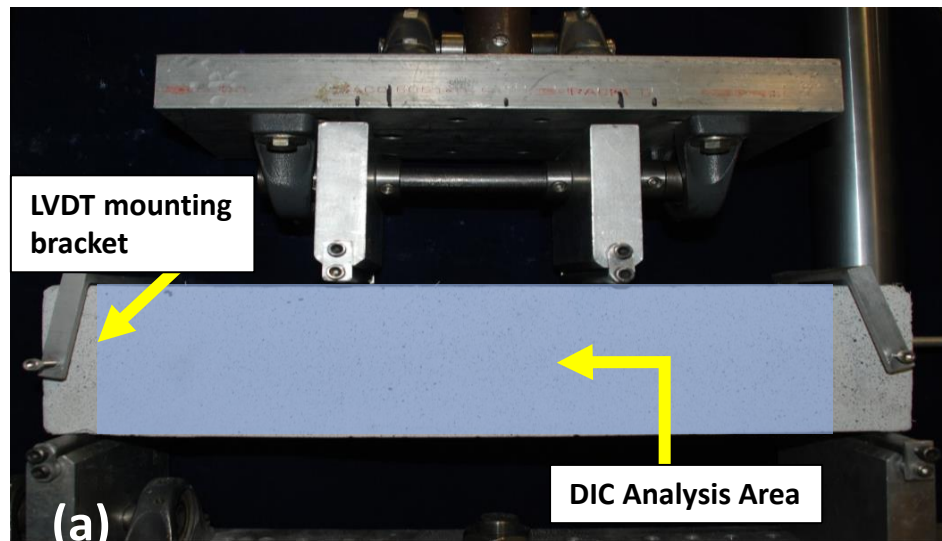


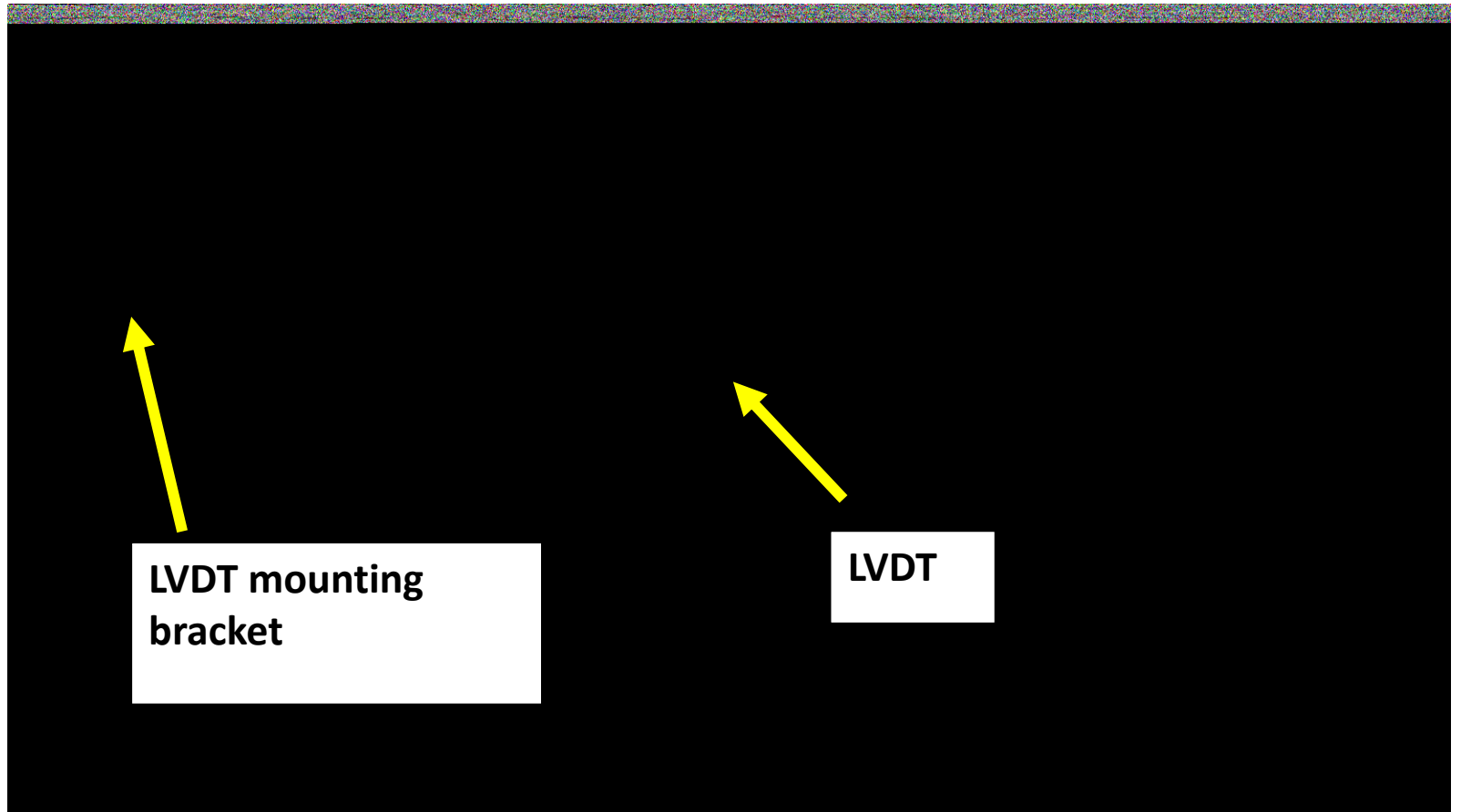
- V.Calard, J.Lamon, "A probabilistic-statistical approach to the ultimate failure of ceramic-matrix composites—part I: experimental investigation of 2D woven SiC/SiC composites," , Composites Science and Technology 62 (2002) 385–393

# Flexural Closed Loop Testing

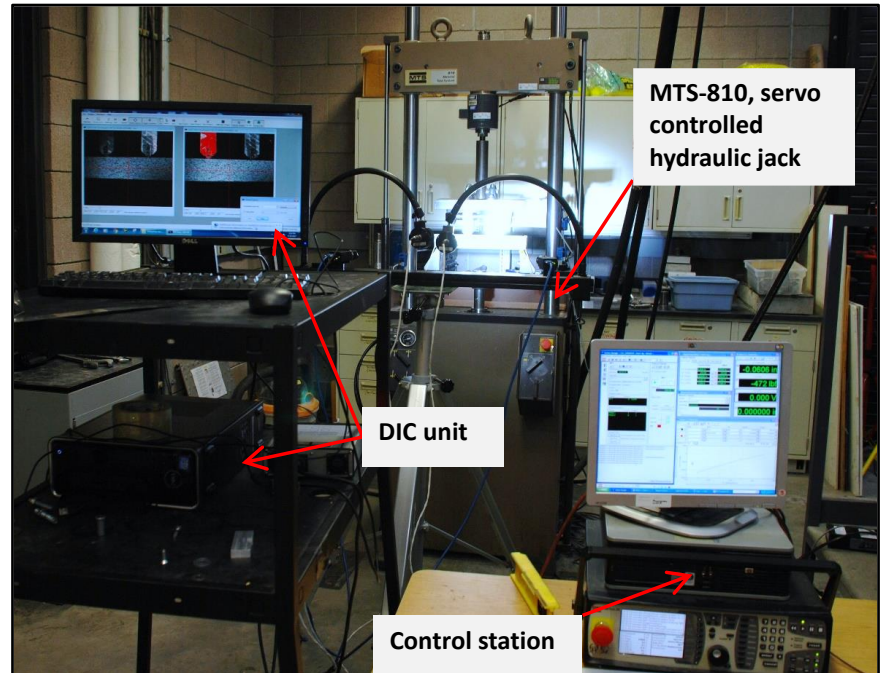
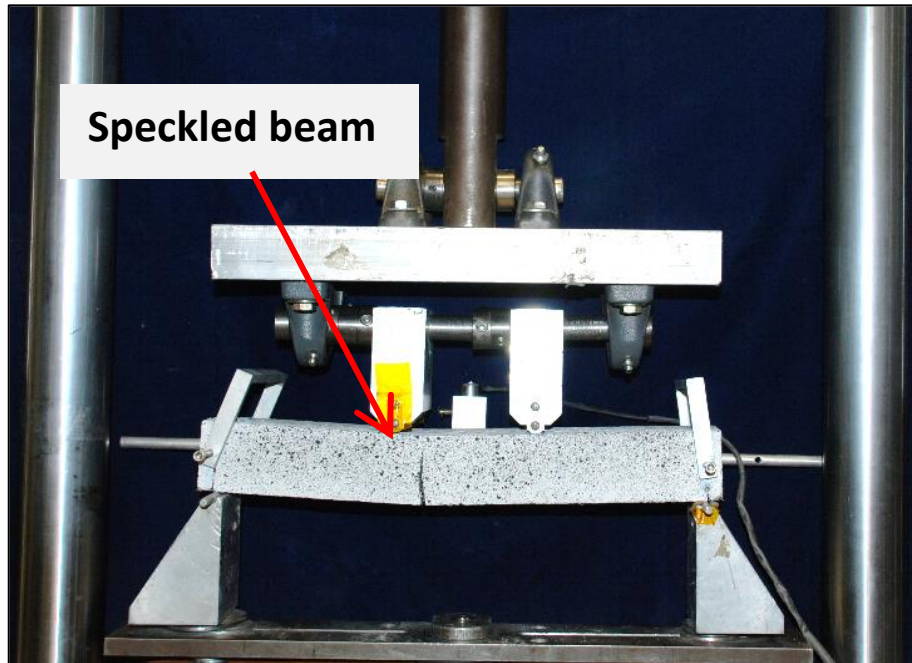
- 28 day flexural response prismatic specimens (2"x2.5"x15") or ASTM C1609.
- Midpoint deflection measured using LVDT placed in the center of the bottom face of the beam
- Strain fields were monitored using digital image correlation (DIC) technique.





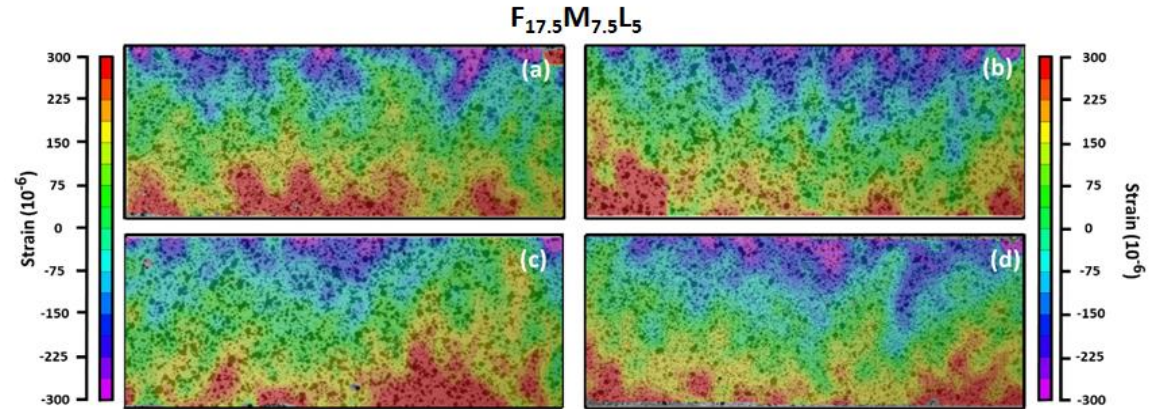


# Flexural closed loop tests

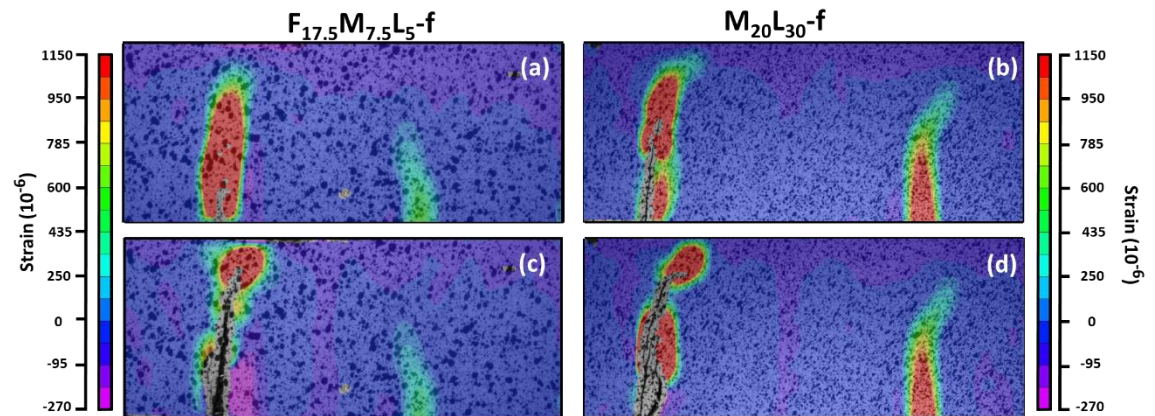


# Flexural Test Results plain and Fiber reinforced

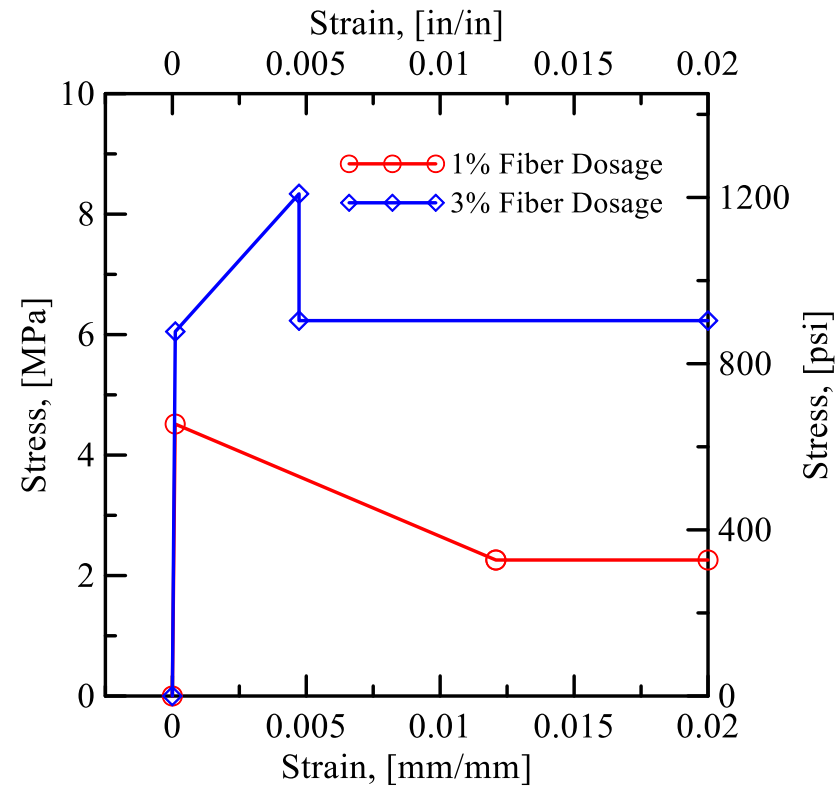
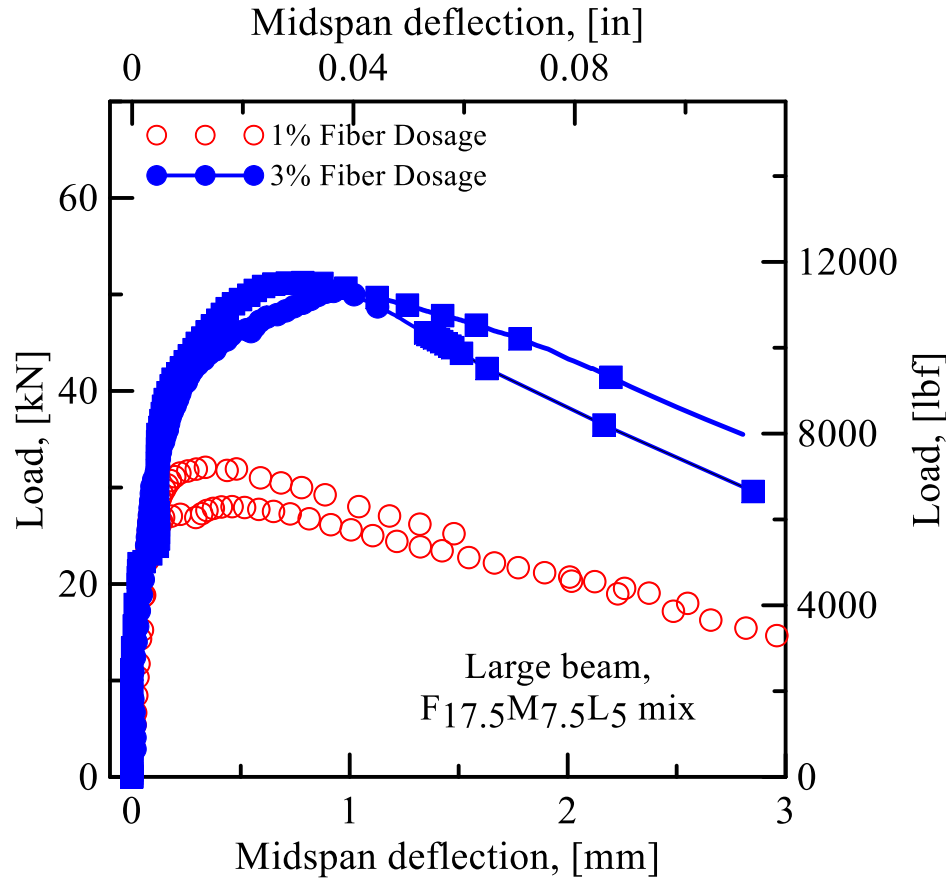
The strain distribution in plain UHPC beams at peak load just prior to unstable failure



Strain fields for fiber reinforced UHPC beam (1% steel fiber) at a post peak stress of (a,b) 90% prepeak and (c,d) 50% postpeak of the peak stress.

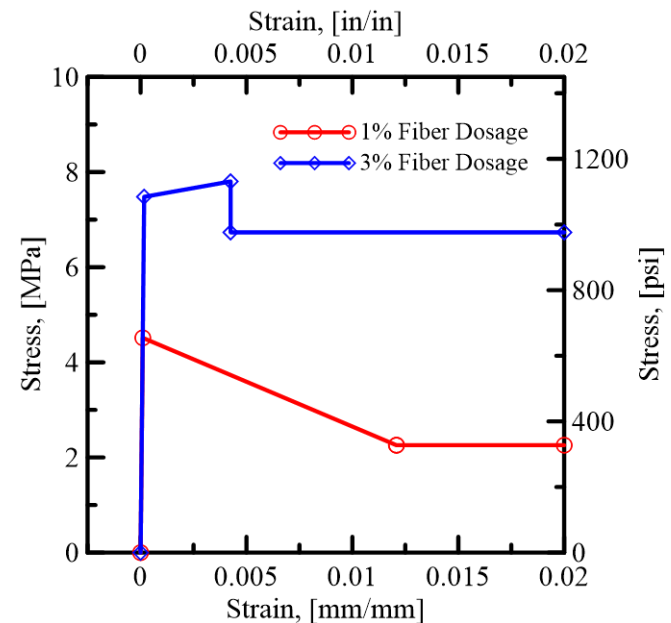
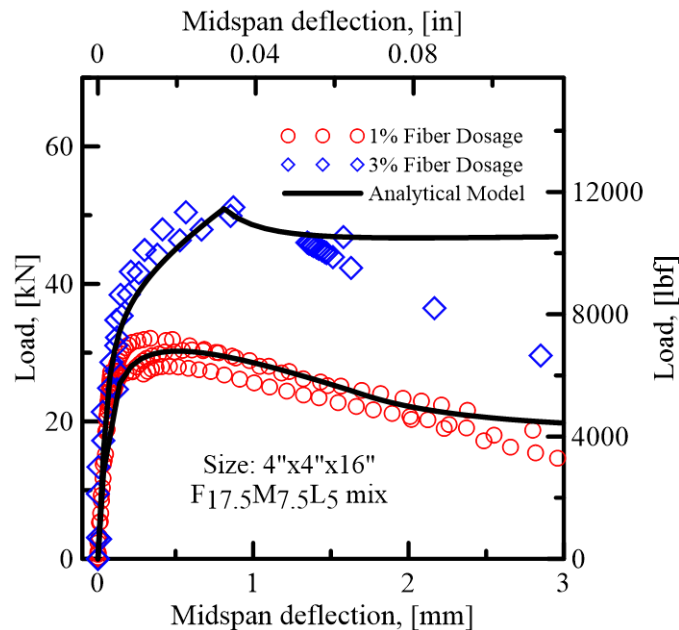


# UHPC with Fibers at 1%- 3% by Volume- Strain Softening and Hardening



# Applications of the UHPC Model

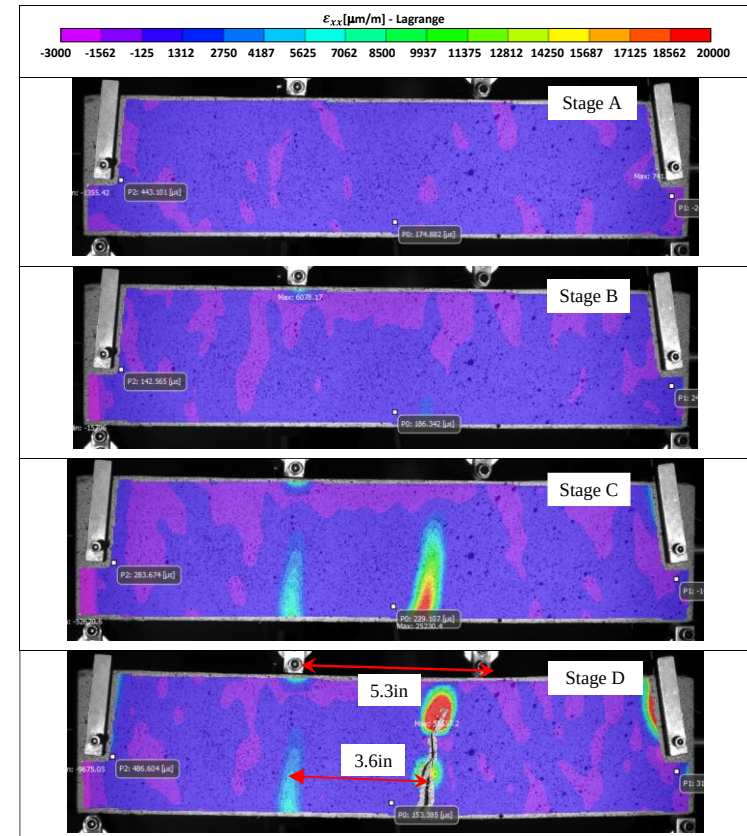
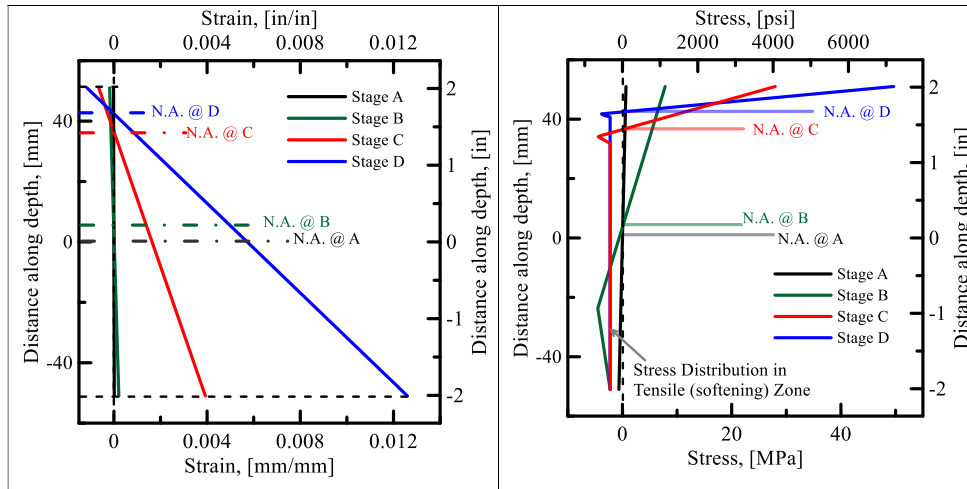
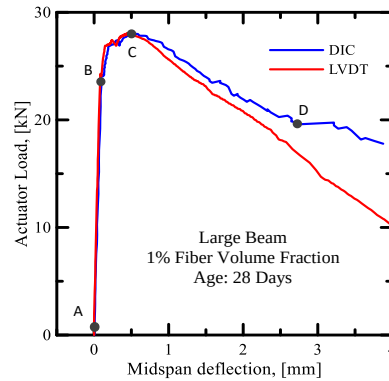
- Compute the Material Property Data Using Inverse Analysis



# Applications of the UHPC Model

- Validation of the DIC Measurements in Flexural Crack Growth

| Stage | Characteristic                              |
|-------|---------------------------------------------|
| A     | Initiation of the deformation               |
| B     | Initiation of the non-linear response       |
| C     | Response to the peak load                   |
| D     | End of the DIC data analysis (beam failure) |



# Mix Design

In the first step of bending tests, two set of tests on two groups of beams, with the same dimension but different mix design, were done.

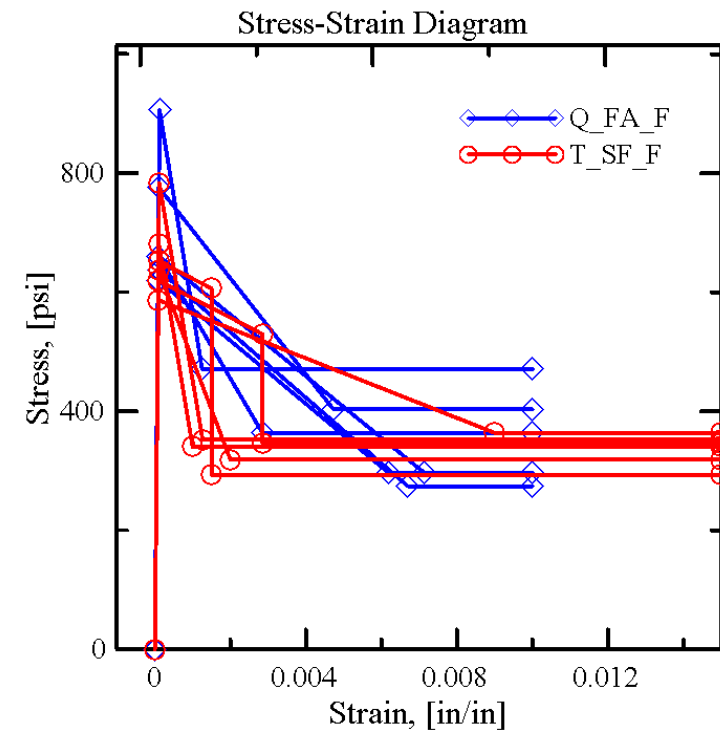
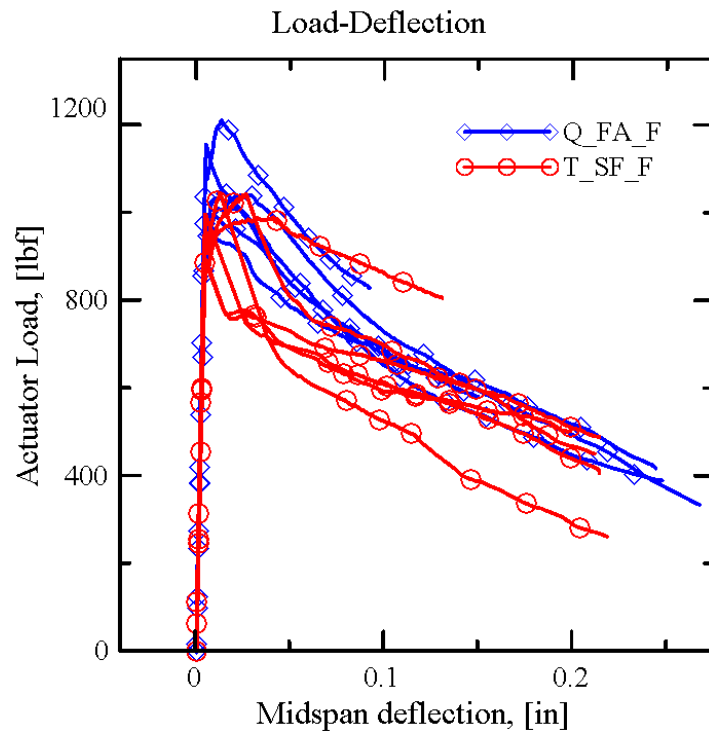
| Material                                                         | Replacement material (% by mass of cement) |                        |
|------------------------------------------------------------------|--------------------------------------------|------------------------|
|                                                                  | Q_FA_F<br>(OPC + F/S + M + L)              | T_SF_F<br>(OPC + M+ L) |
| <b>Fly Ash (F) / Slag (S)</b>                                    | 17.5                                       | 0                      |
| <b>Metakaolin (K)</b>                                            | 0                                          | 0                      |
| <b>Microsilica (M)</b>                                           | 7.5                                        | 20                     |
| <b>Limestone (L); d50 of 1.5 or 3 <math>\mu\text{m}^*</math></b> | 5b, 5c                                     | 30                     |
| <b>Steel Fiber</b>                                               | 1                                          | 1                      |



| Sample ID     | Q_FA (MPa)         | QFA-F (MPa)        | T_SF (MPa)         | T_SF_F (MPa)          |
|---------------|--------------------|--------------------|--------------------|-----------------------|
| <b>Day 3</b>  | 120 ( $\pm 1.90$ ) | 108 ( $\pm 1.43$ ) | 102 ( $\pm 2.70$ ) | 100.04 ( $\pm 0.98$ ) |
| <b>Day 7</b>  | 130 ( $\pm 1.27$ ) | 123 ( $\pm 3.73$ ) | 117 ( $\pm 4.76$ ) | 110 ( $\pm 3.01$ )    |
| <b>Day 28</b> | 144 ( $\pm 4.28$ ) | 149 ( $\pm 2.97$ ) | 127 ( $\pm 7.38$ ) | 133 ( $\pm 1.69$ )    |

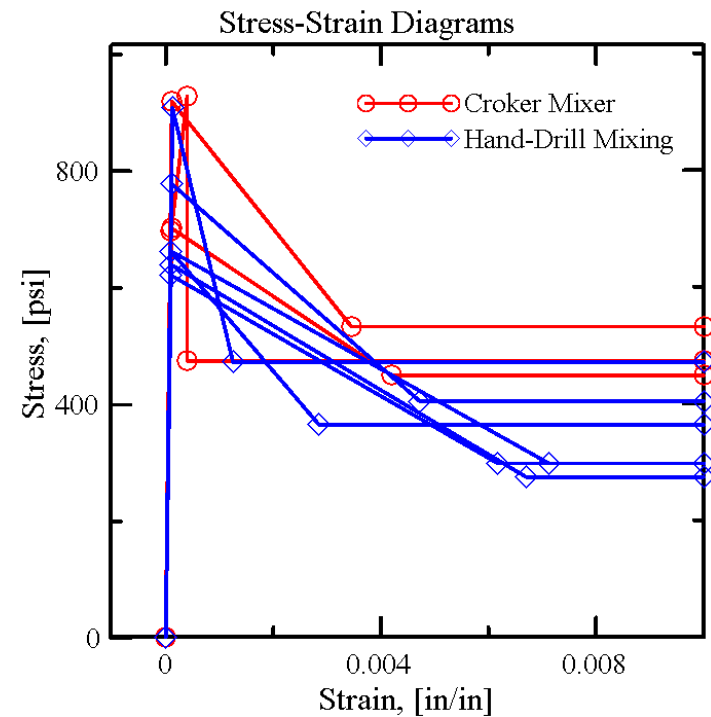
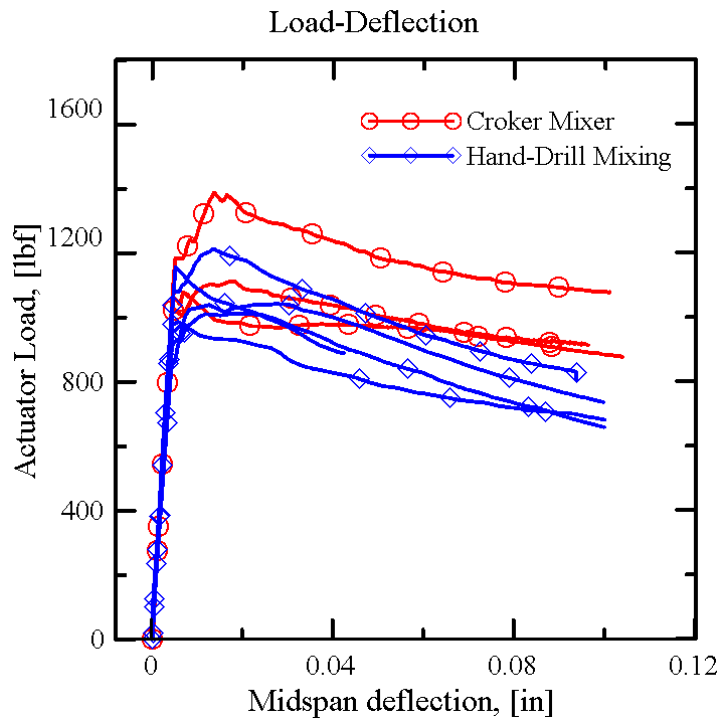
Q\_FA series showed better compressive performance

# Effect of mix design, Multiple Samples



- Small beams (2"X2.5"X14")
- 28 days age

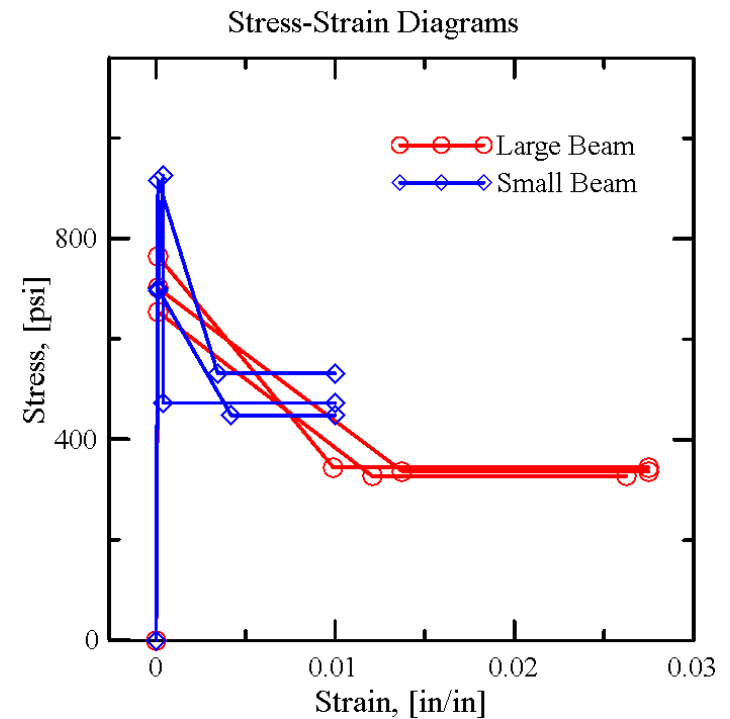
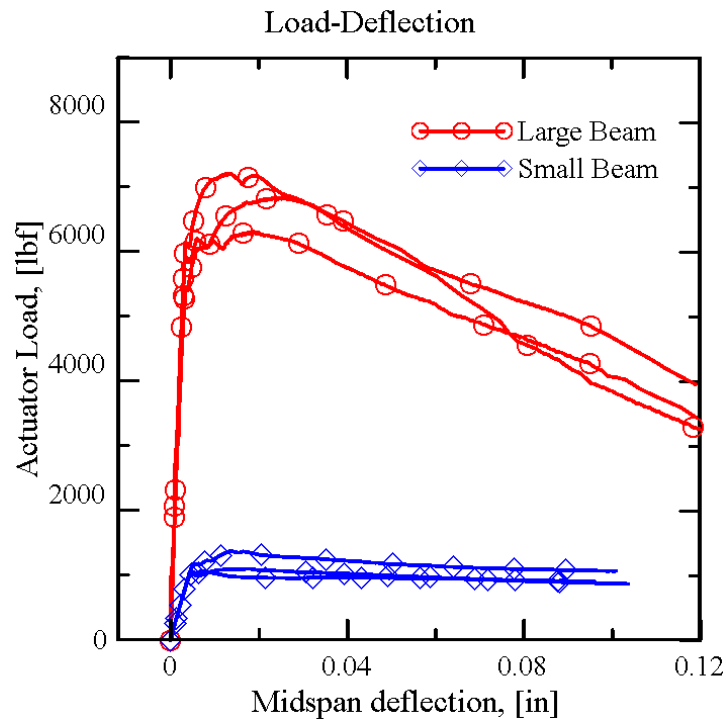
# Effect of Mixer Type



- Small beams (2"X2.5"X14")
- 28 days age
- Q\_FA\_F samples

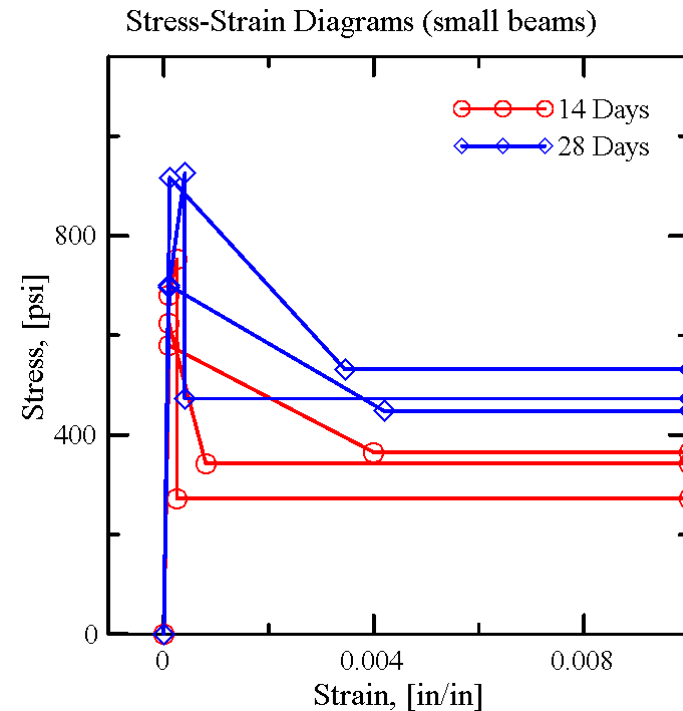
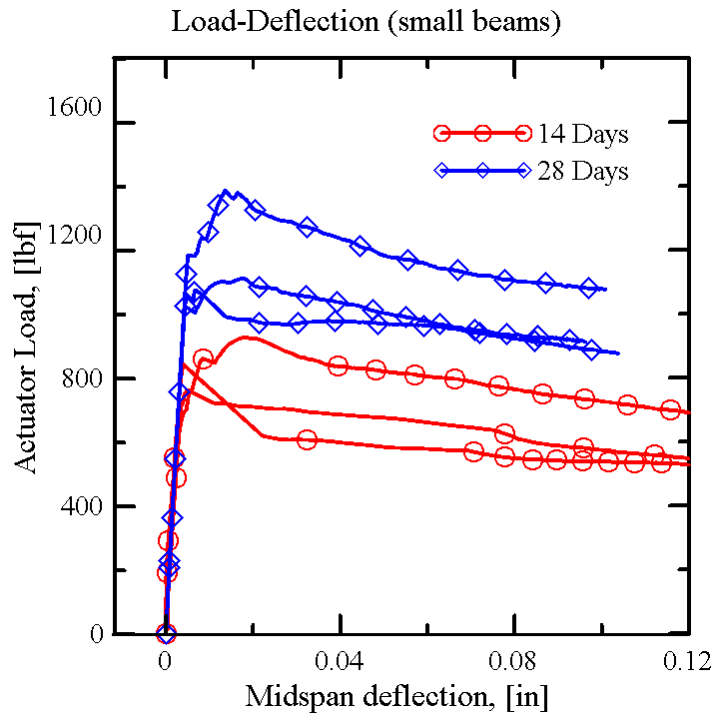
Using Croker mixer as the mixing method gives higher post-crack performance.

# Effect of Sample Size



- Small beams (2"X2.5"X14") and large beams (4"X4"X16")
- Small beams show lower strain capacity, but higher stress capacity.

# Effect of Curing

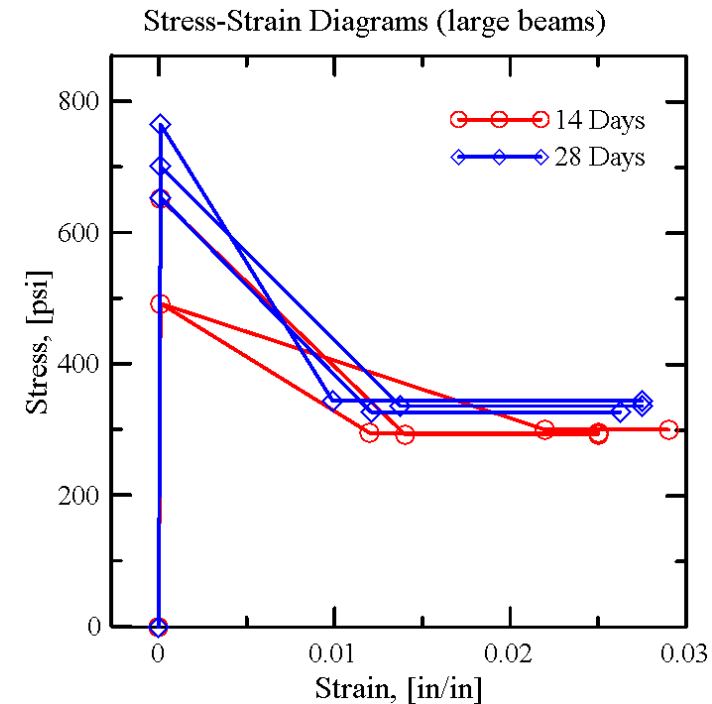
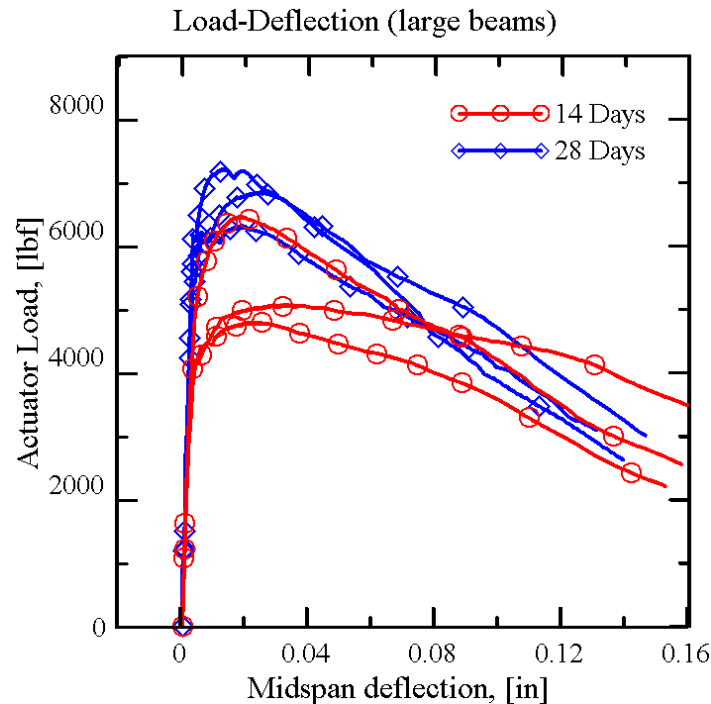


28 days samples show around 40% higher loading (stress) capacity  
Strain capacity does not show any significant change as age changes.

## Effect of the age on large beams:

- Large beams (4"X4"X16")
- 14 days and 28 days age
- Q\_FA\_F samples

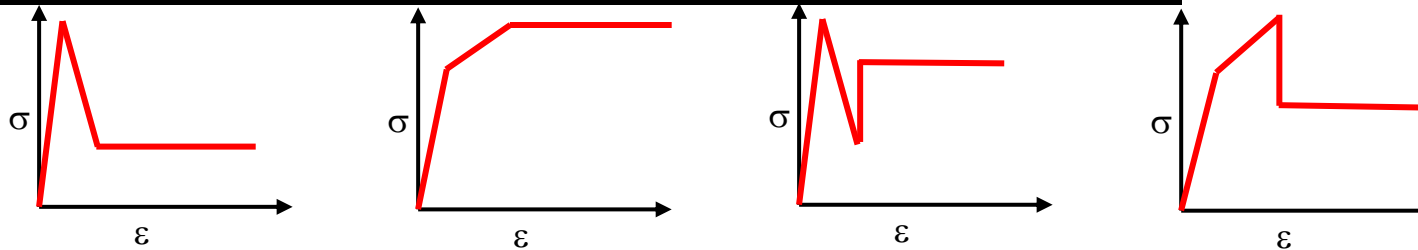
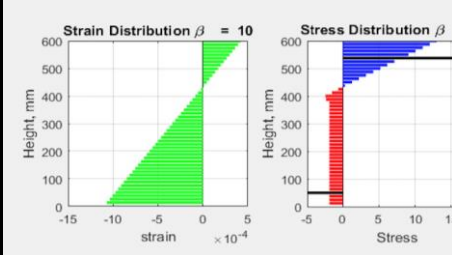
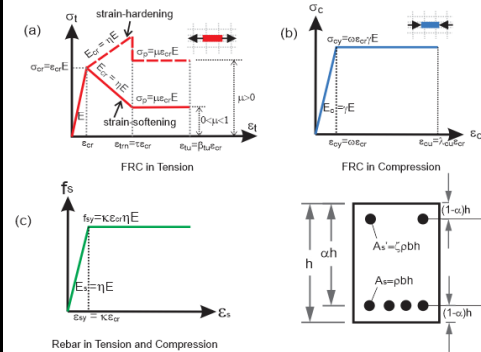
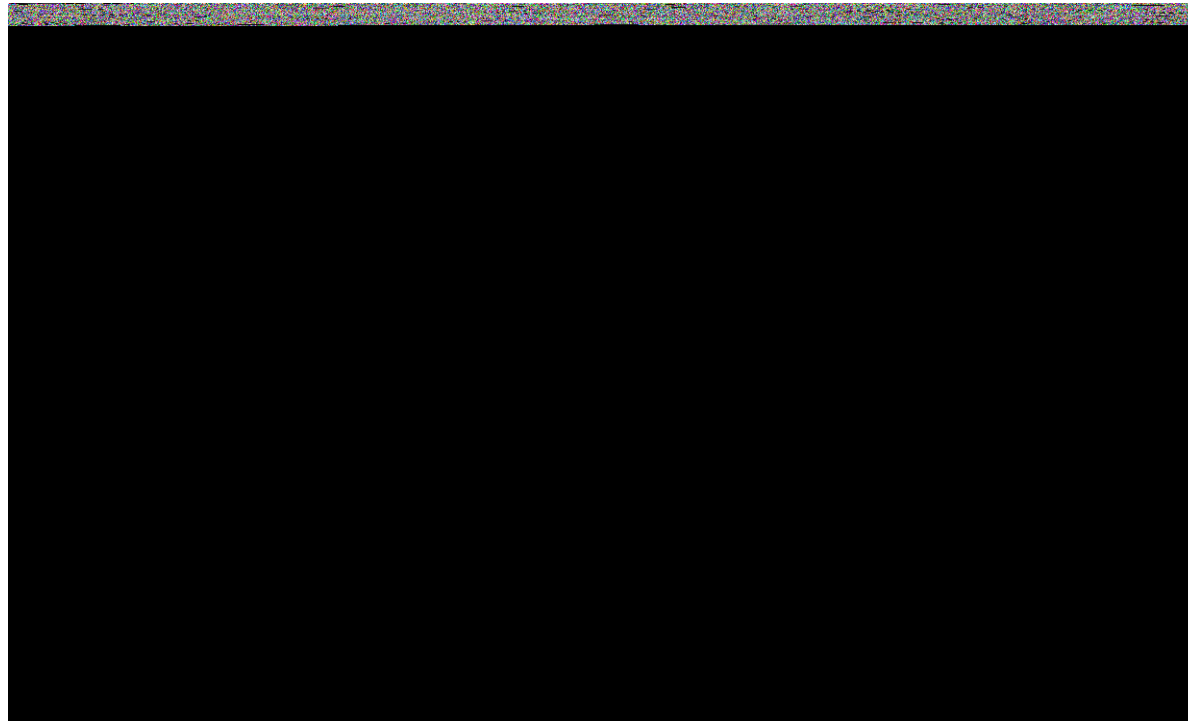
28 days samples show around 23% higher loading (stress) capacity, however their strain capacity does not show any significant change as age changes.



# Matlab GUI and Excel applications

HRC Back-Calculation software (Matlab)

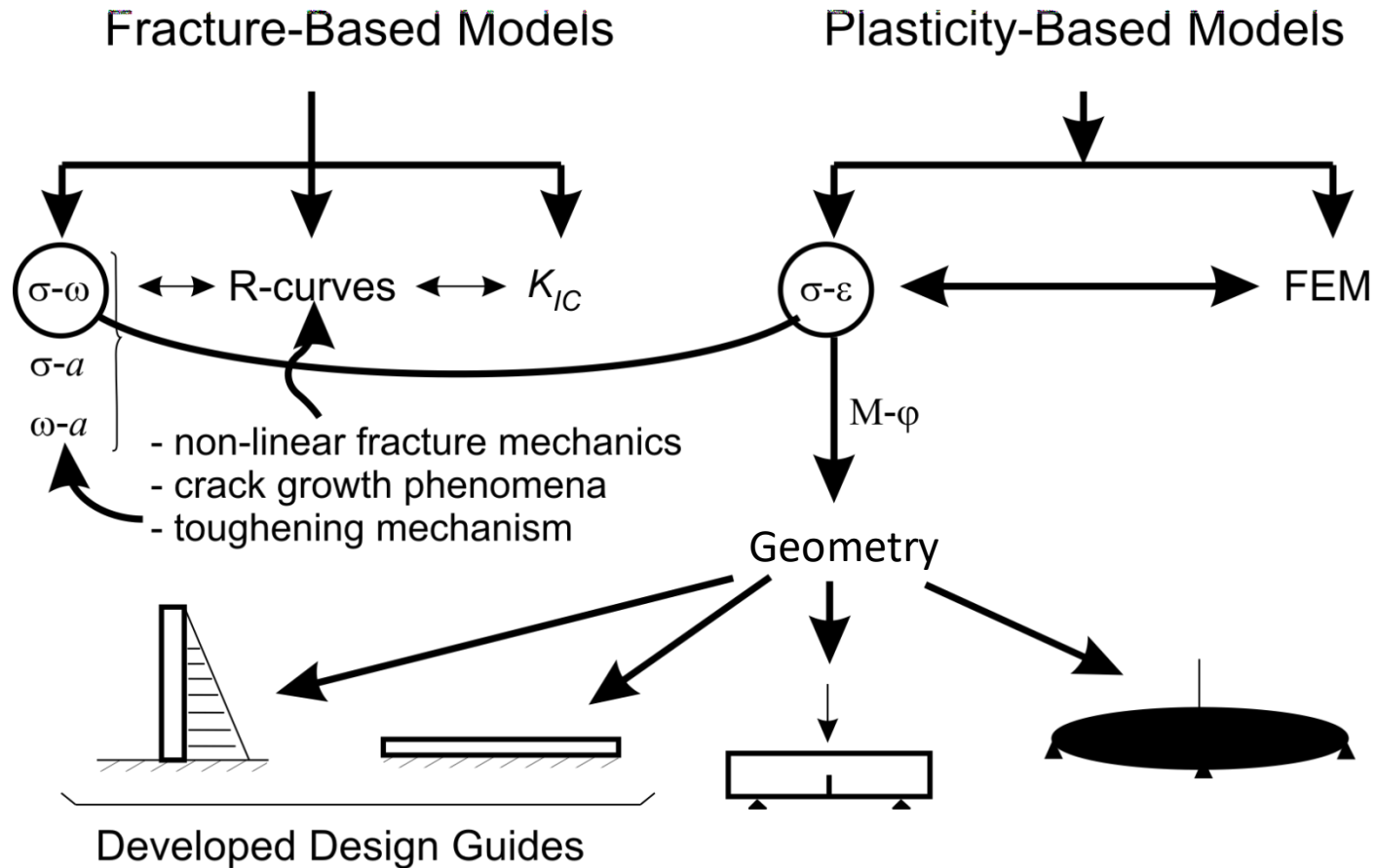
Output will produce 12 plots with an output file



# Supporting software and guides

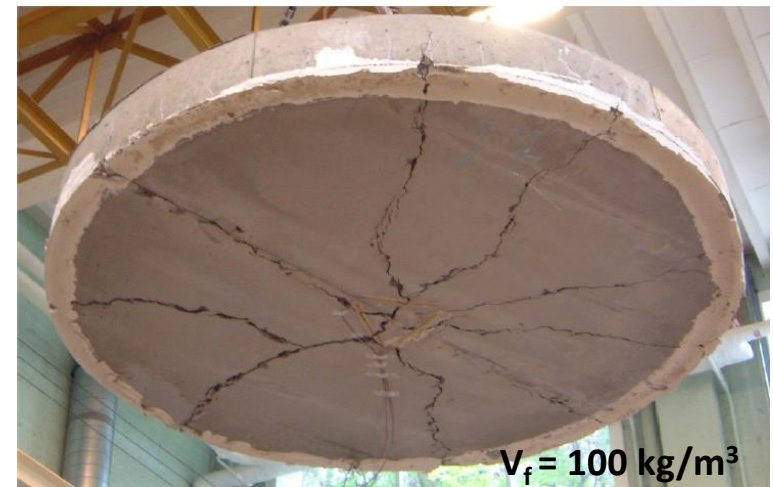
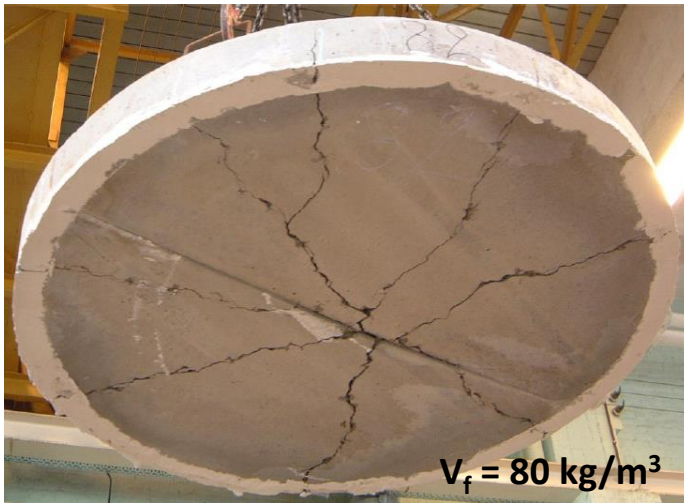
- An excel spreadsheet with associated uses manual that computes the material properties for three- and four-point bending samples
- Versions for Hybrid Design, (with tension reinforcement ) compression P-M diagram are also available for download
- ACI 544 Committee web page [https://www.concrete.org/committees/directoryofcommittees/acommitteehome.aspx?committee\\_code=0000544-00](https://www.concrete.org/committees/directoryofcommittees/acommitteehome.aspx?committee_code=0000544-00)
- ResearchGate Data Barzin Mobasher [https://www.researchgate.net/profile/Barzin\\_Mobasher](https://www.researchgate.net/profile/Barzin_Mobasher)
- A Spreadsheet-Based Inverse Analysis Procedure for Flexural Specimens – Strain Softening/Hardening Behavior
- DOI 10.13140/RG.2.1.4409.2882
- [https://www.researchgate.net/publication/283722776\\_A\\_Spreadsheet-Based\\_Inverse\\_Analysis\\_Procedure\\_for\\_Flexural\\_Specimens - Strain\\_SofteningHardening\\_Behavior](https://www.researchgate.net/publication/283722776_A_Spreadsheet-Based_Inverse_Analysis_Procedure_for_Flexural_Specimens_-_Strain_SofteningHardening_Behavior)

# Fracture and Plasticity models

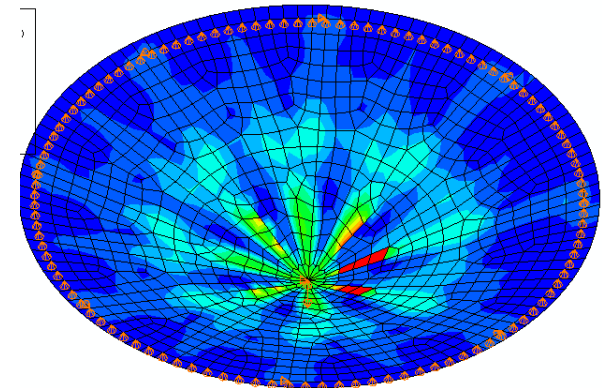
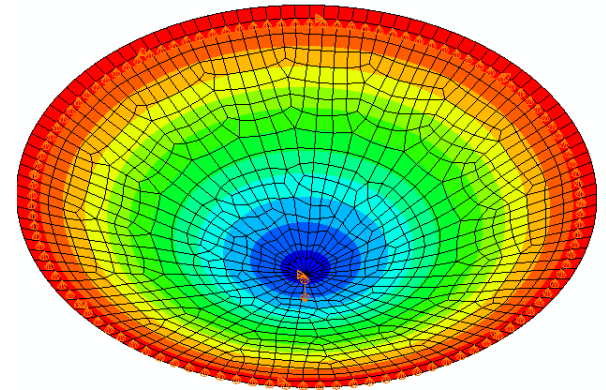
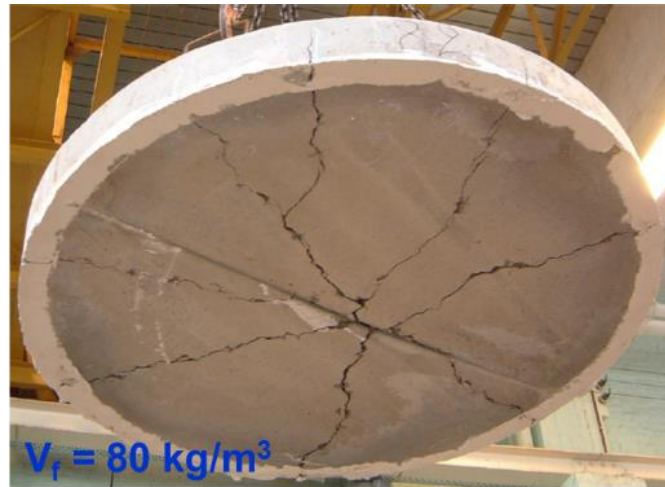
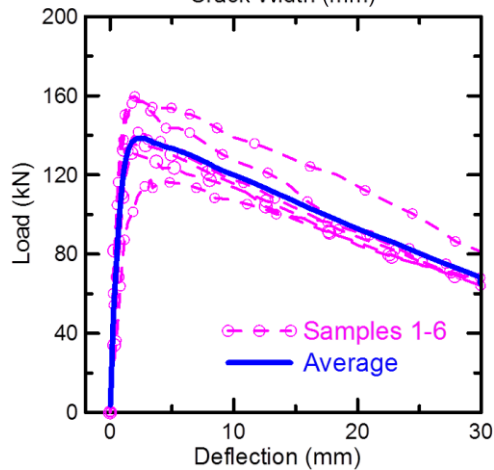
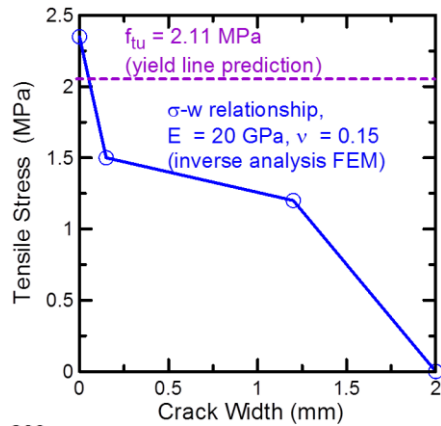


# Material Ductility vs. Structural Ductility

- initiation of the first Full structural crack does not lead to the total failure of the specimen due to the Structural redundancy
- Test setup- displacement controlled, continuous support, center point load
- Dimensions-  $D, t = 1500, 150 \text{ mm}$



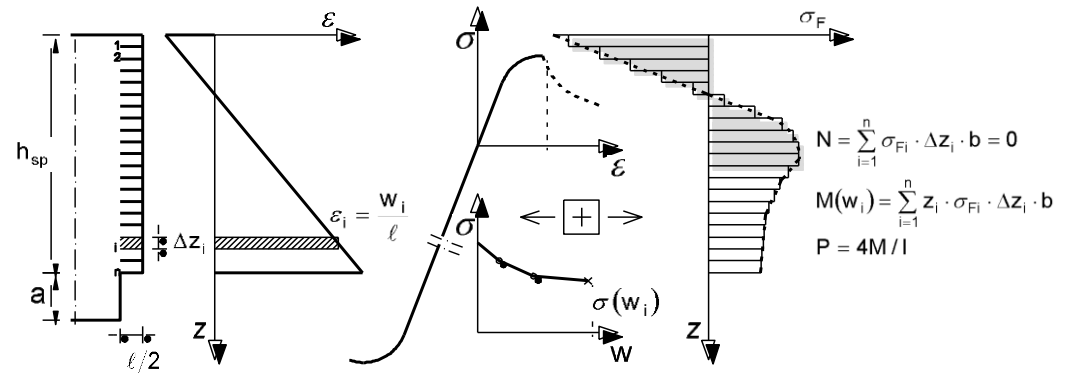
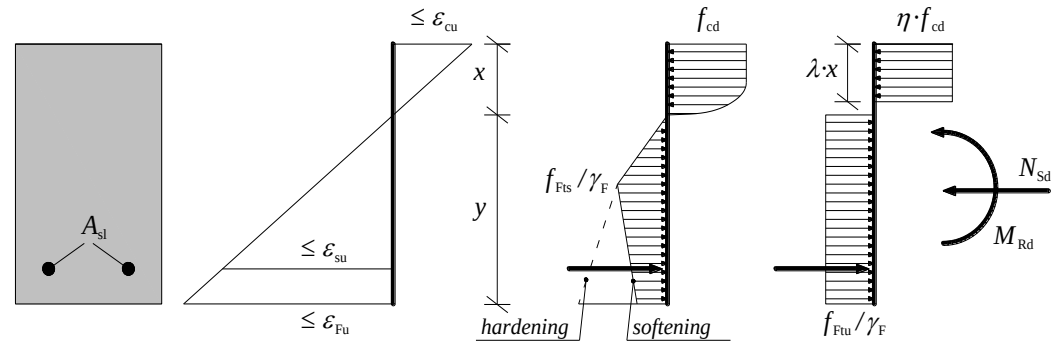
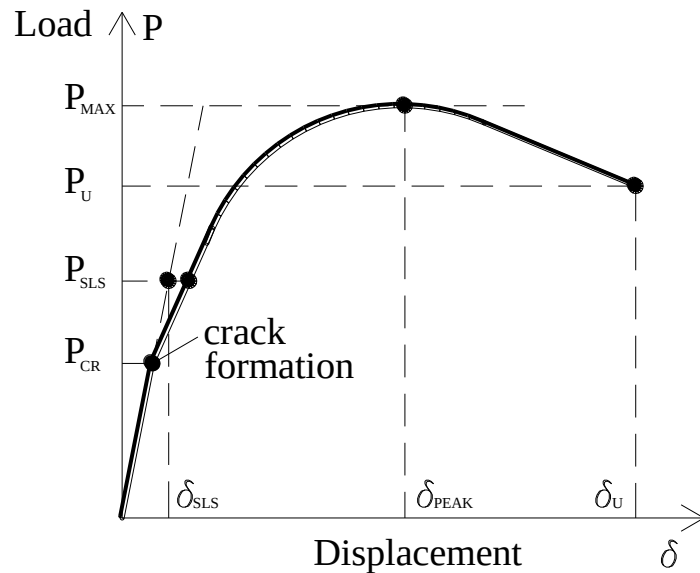
# Round Panel Continuous Support Specimens- multiple cracking in flexure strain softening leads to deflection hardening



### 3. Model Code- Design guides

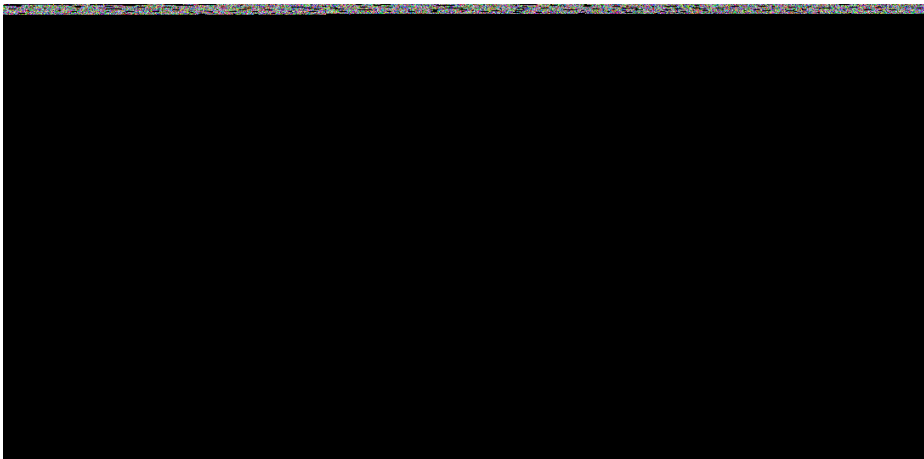
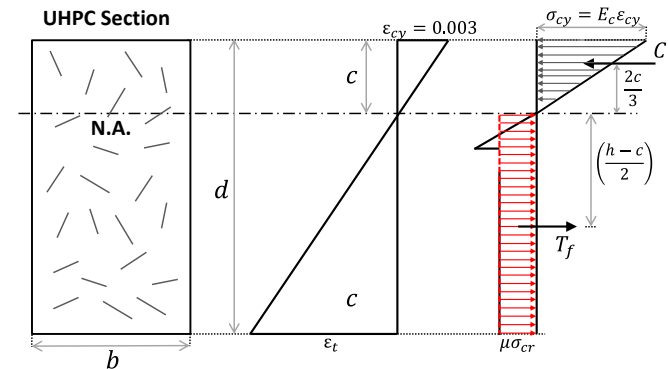
#### Design Philosophy for Flexure

load (P) – displacement



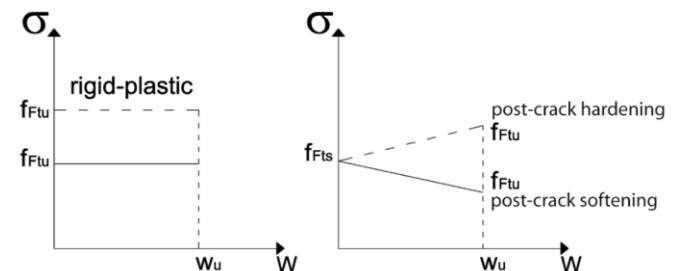
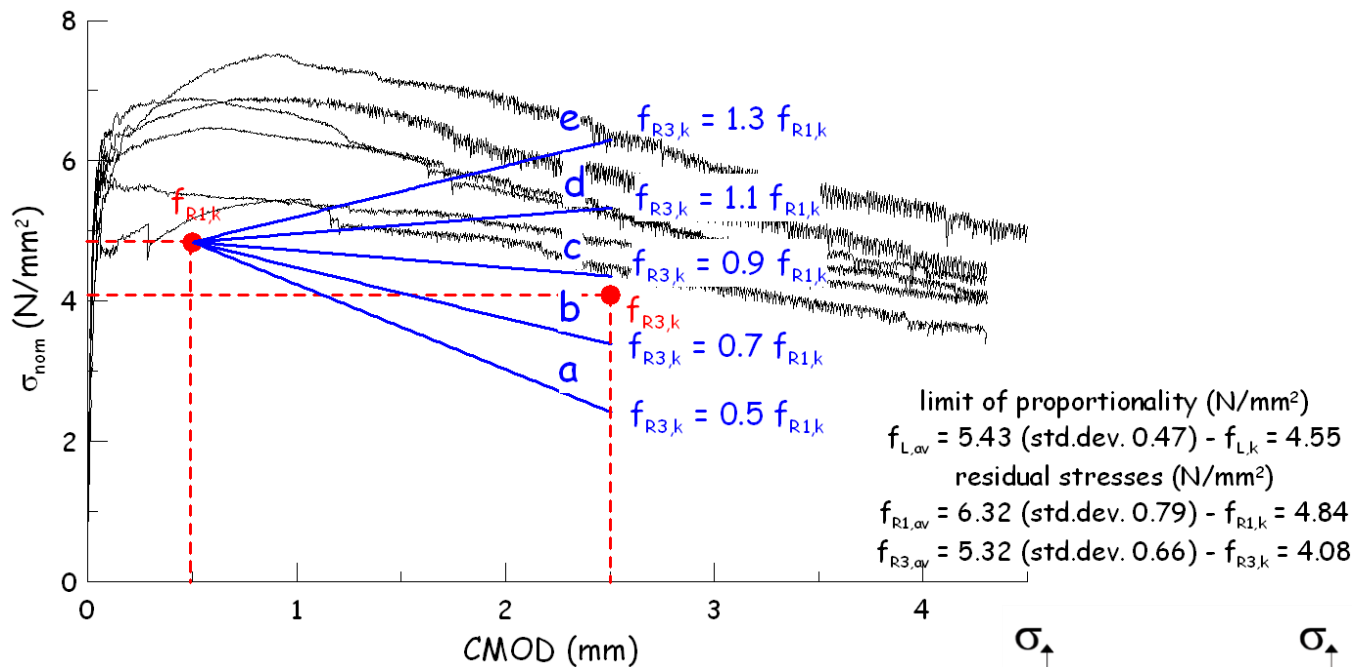
# Analysis of Flexural UHPC Members

- The classic ACI approach, based on Whitney rectangle concept
- Ultimate limit state (ULS) approach
- Constitutive analytical approach, used for the serviceability limit state (SLS) approach.



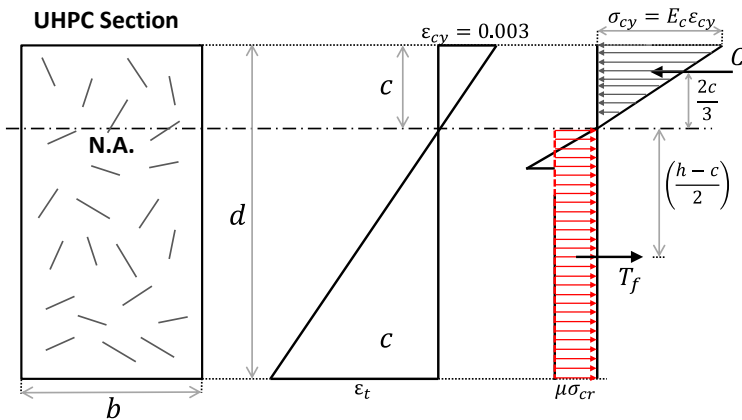
# Model Code Approach

- The procedure developed by EN 14651 requires running a 3-point bending test on a notched beam, collecting the load vs. crack opening response.

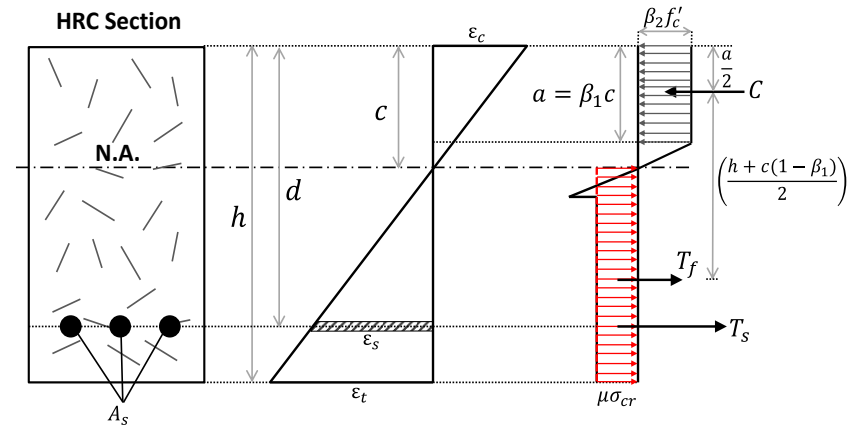


# Simplified Approach for Incorporation of Fibers in Flexural Model

- Classic approach based on Whitney rectangle concept for estimating the bending capacity of a plain UHPC beam



- Fiber Reinforced Concrete (FRC)
- Compressive failure is unlikely
- Maximum compressive stress value of  $\sigma_{cy} = E_c \epsilon_{cy}$
- Tensile response of the section is replaced with a uniform distributed loading  $\mu\sigma_{cr}$

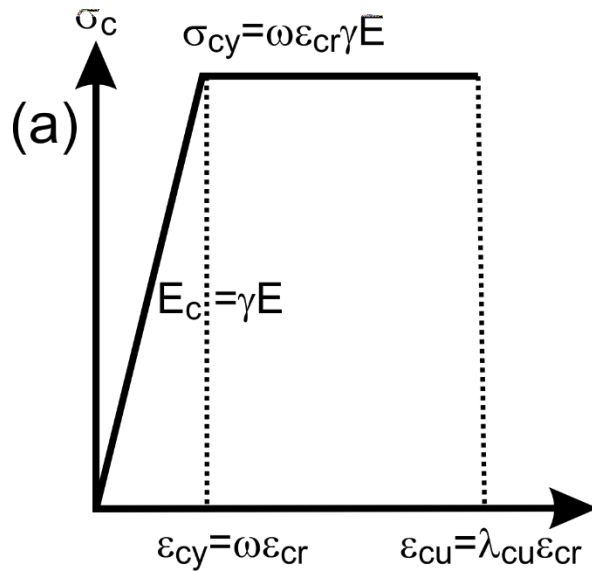


- Hybrid Reinforced Concrete (HRC)
- Plastic compressive failure is likely
- Uniform compressive tension of  $\beta_2 f'_c$
- Tensile response of the section is replaced with a uniform distributed loading  $\mu\sigma_{cr}$

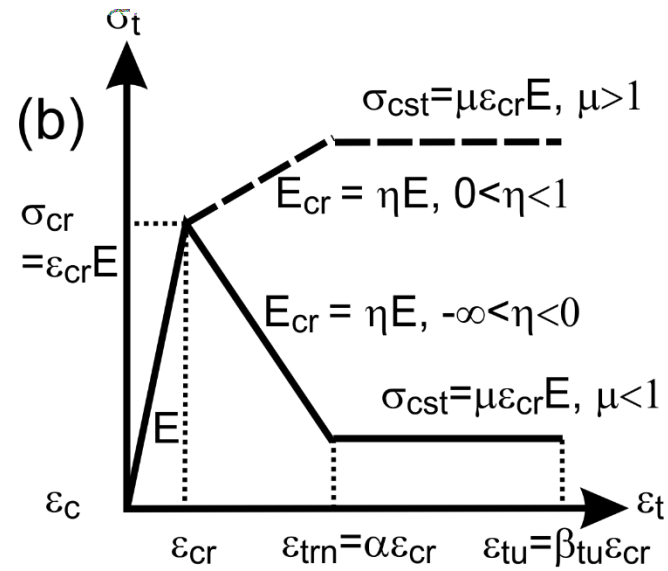
# Flexural Modeling: Stress-strain Model

- Material parameters are described as a multiple of the first cracking tensile strain ( $\varepsilon_{cr}$ ) and tensile modulus ( $E$ )

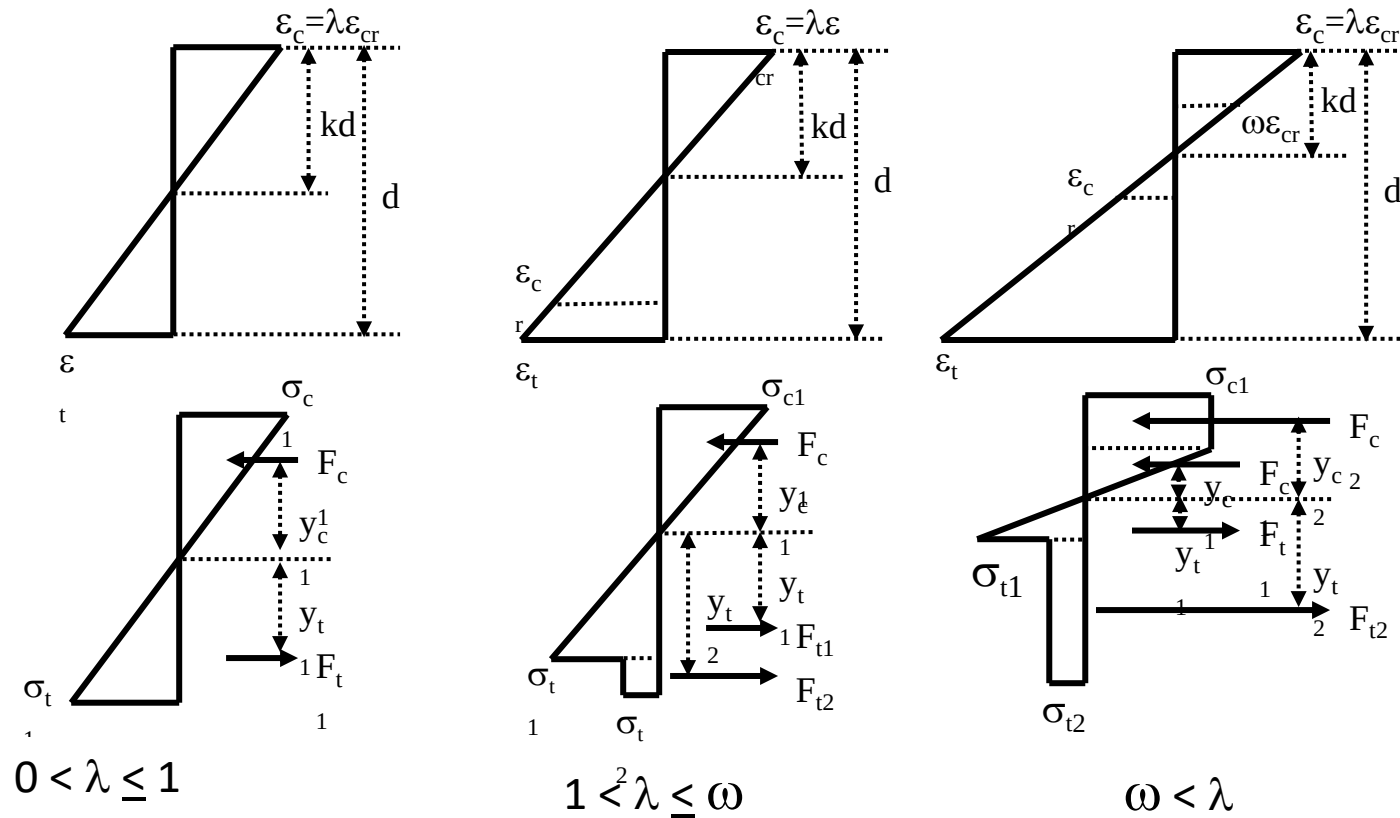
## Compression model



## Tension model

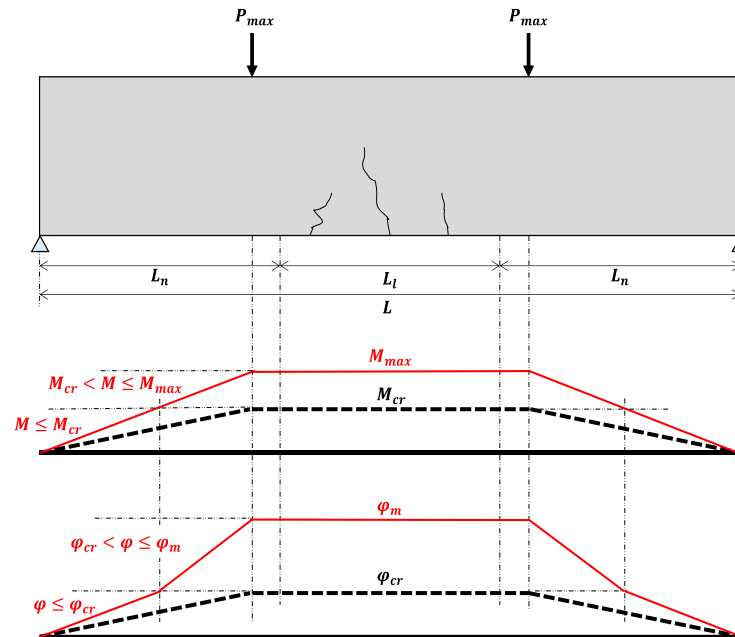
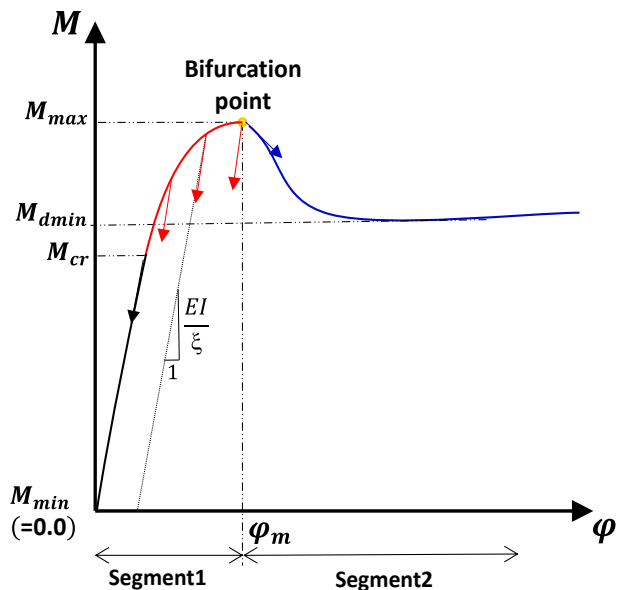


# Stress and Strain Distribution



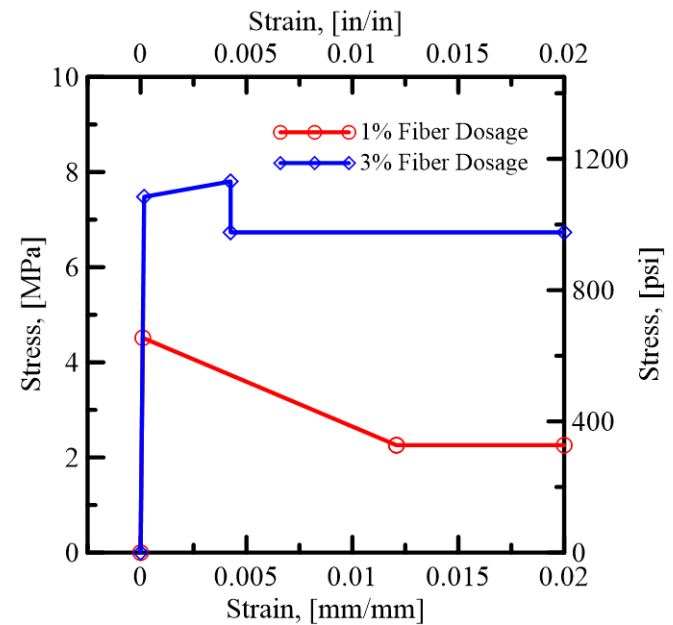
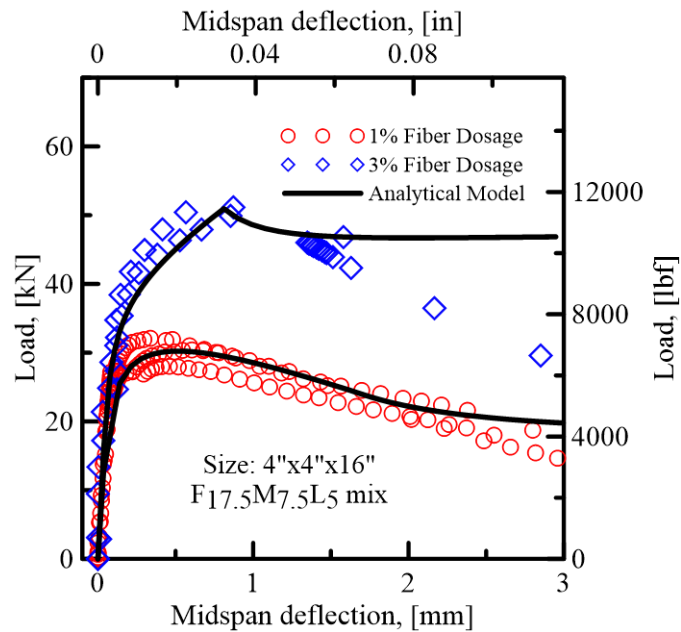
# Load Deflection Computation

- Moment-curvature diagrams are calculated for the sections
- To have a load-deflection response for a beam element, it is necessary to have the moment-curvature response for various sections (elements) along the beam
- Equilibrium is used to obtain support reactions and moment distribution
- The slope and deflection distributions are obtained by applying the slope-deflection method



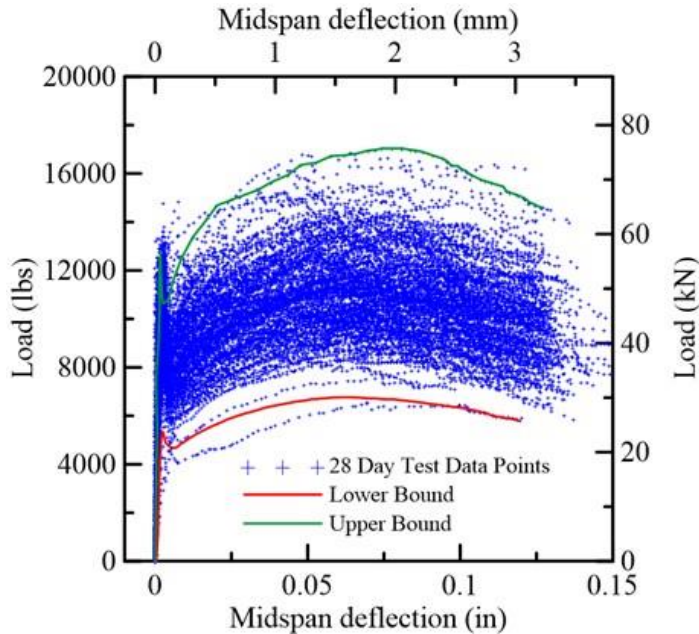
# Applications of the UHPC Model

- Compute the Material Property Data Using Inverse Analysis

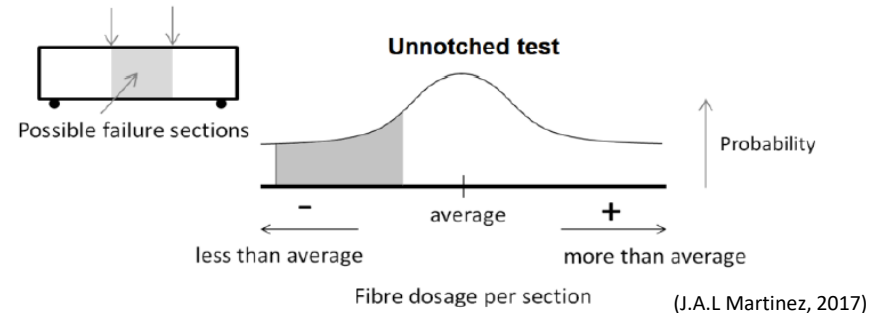
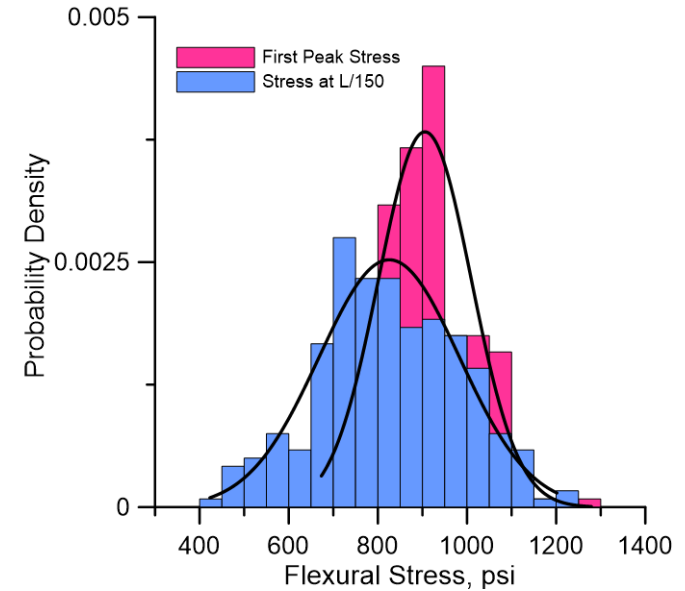


# Study of Variation in Test results

# ASTM C1609 Data: What we've learned so far



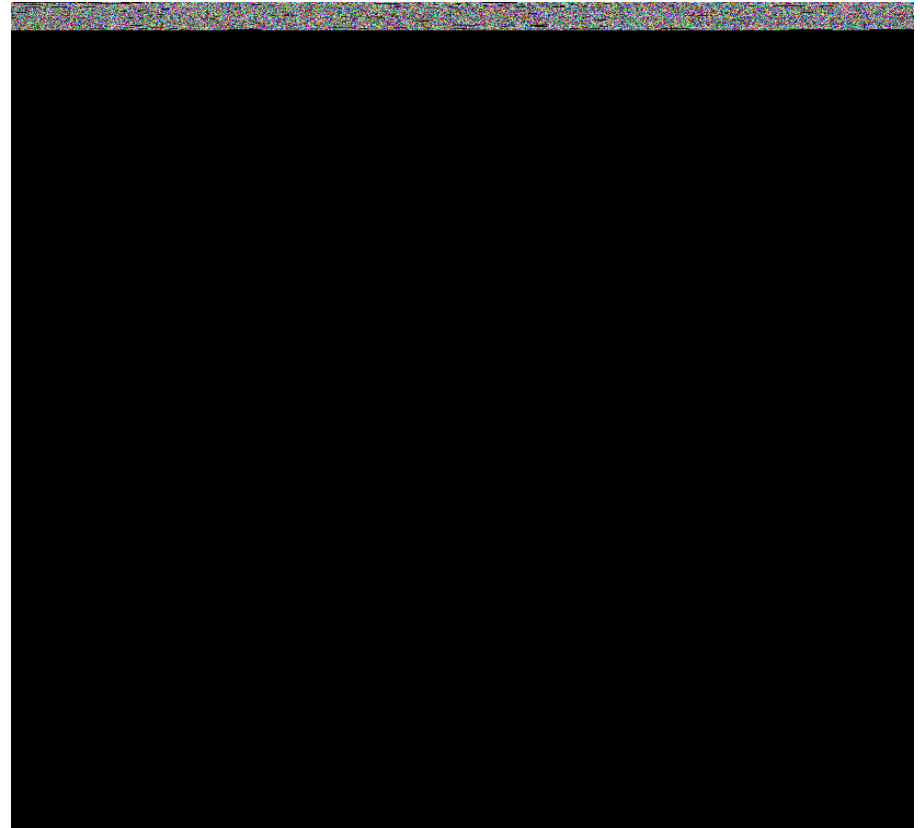
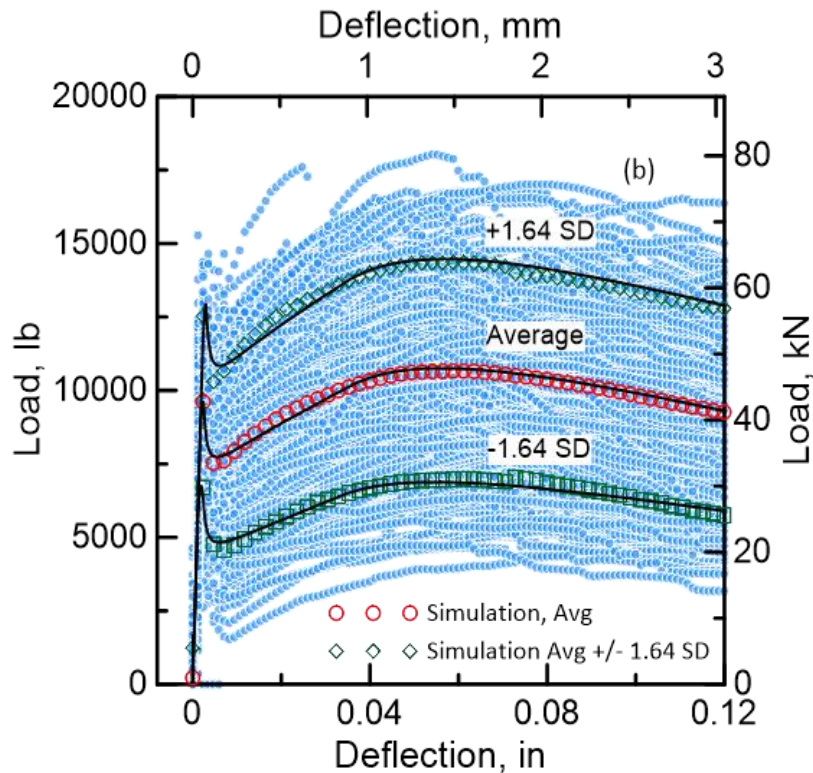
- Lower Variability in MOR compared to other parameters as crack will form at the weakest link inside the constant moment zone.
- Higher stress at MOR
- Higher MOR can reduce overall response of the fiber

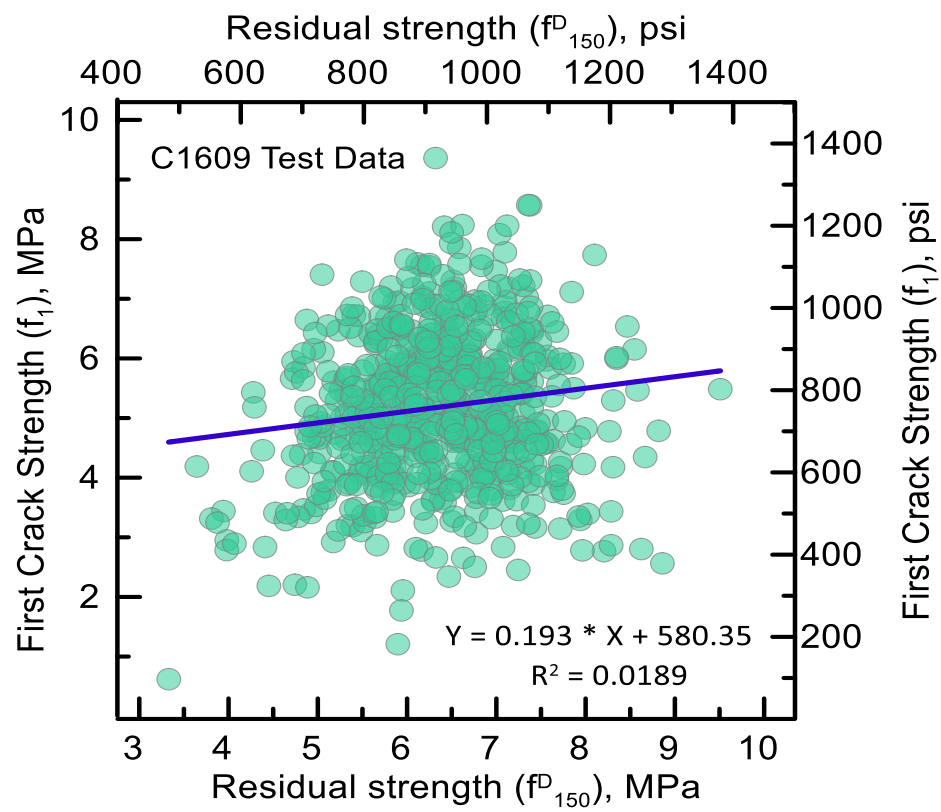
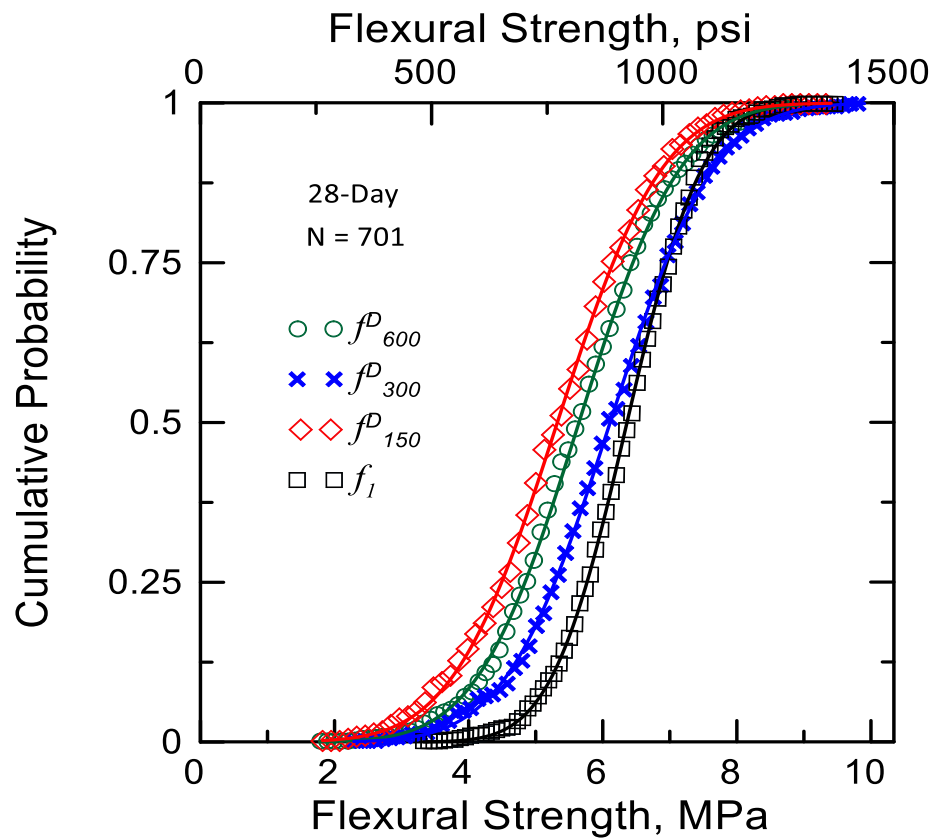


weakest link will have less fiber dosage than average

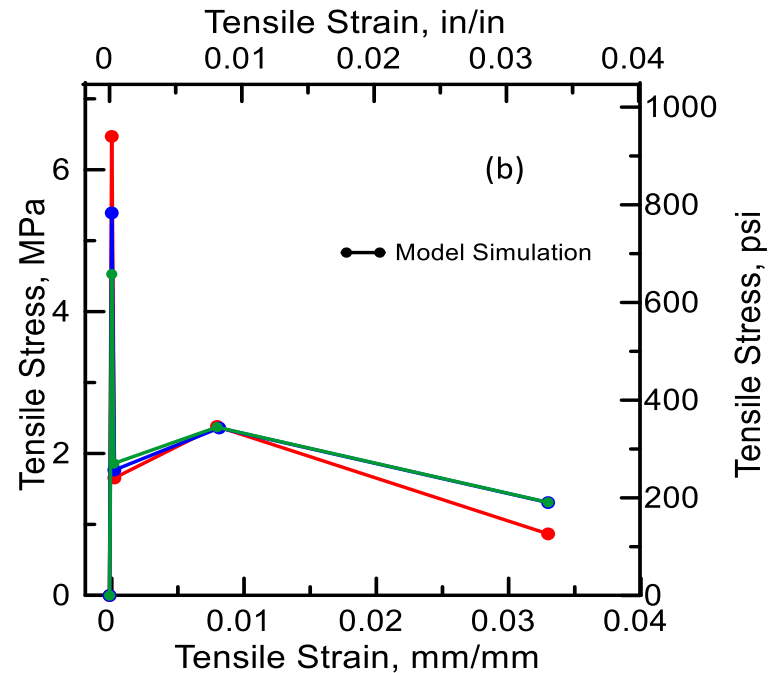
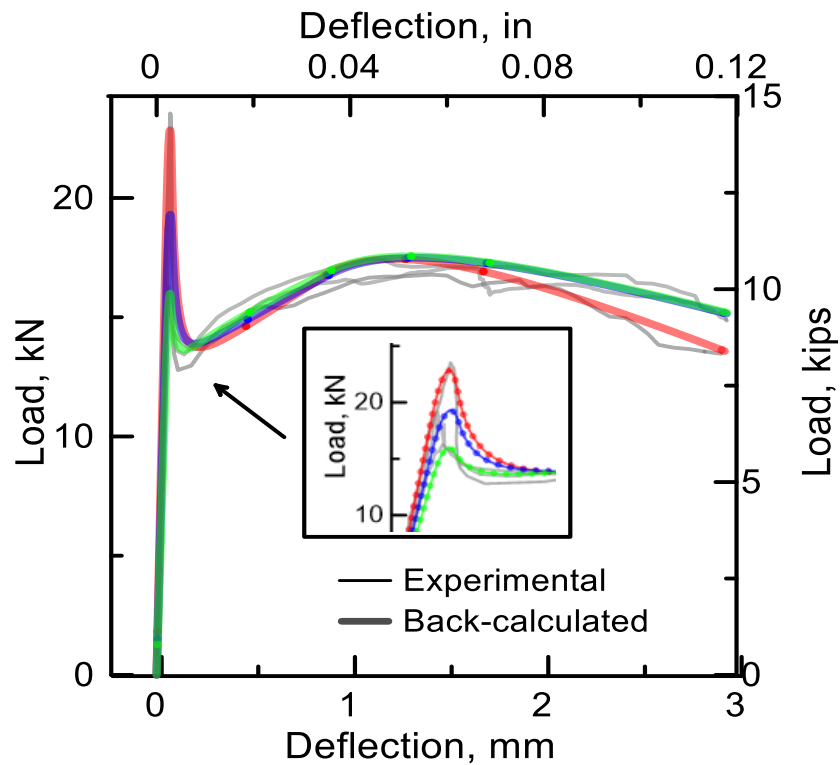
(J.A.L Martinez, 2017)

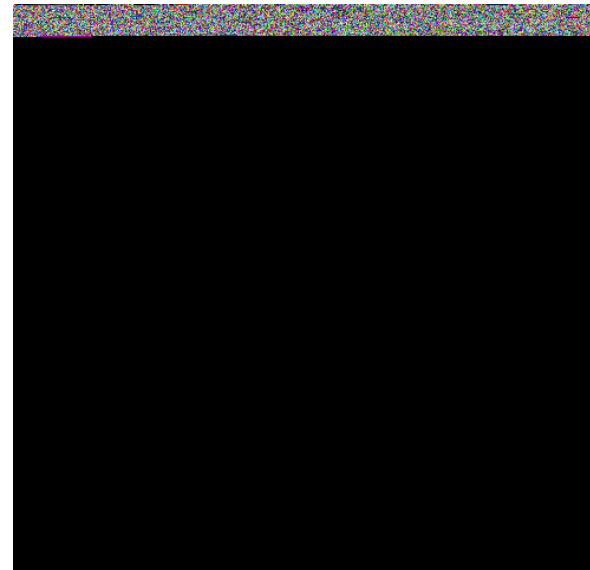
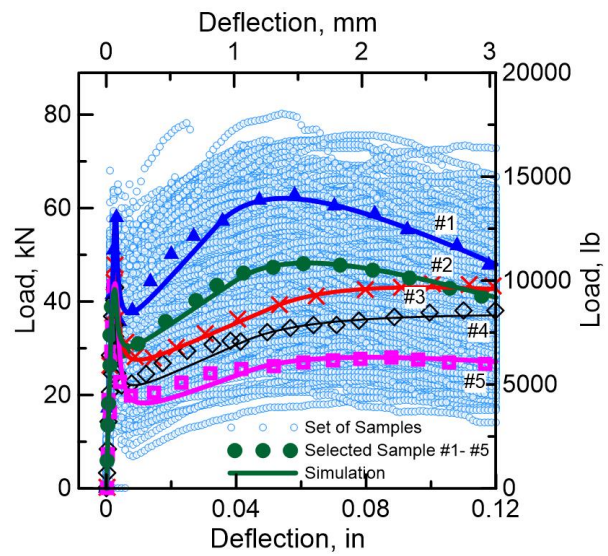
# Variability in the test results using Production data- Precast tunnel lining 45kg/m<sup>3</sup> SFRC



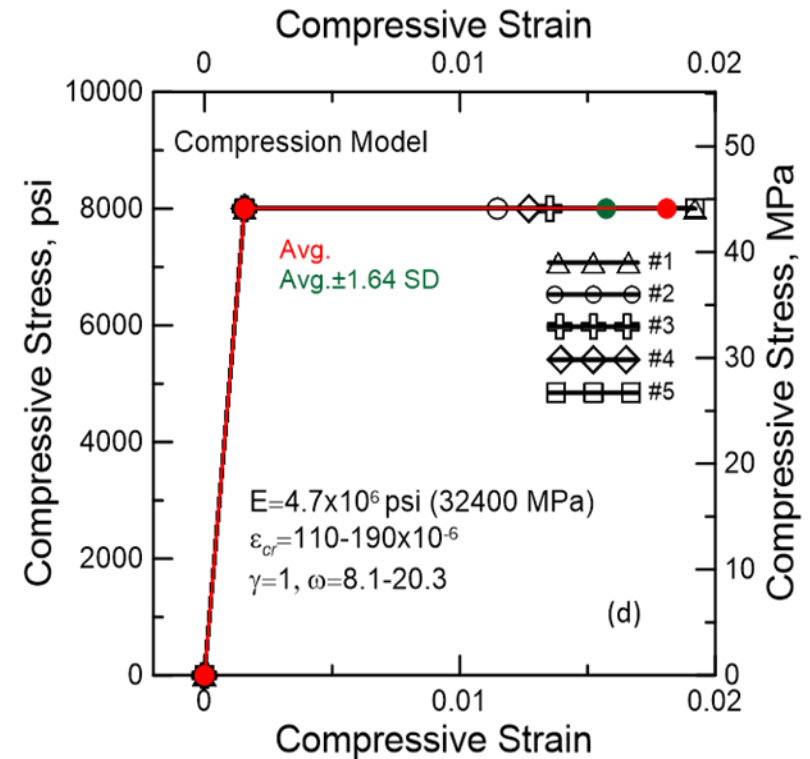
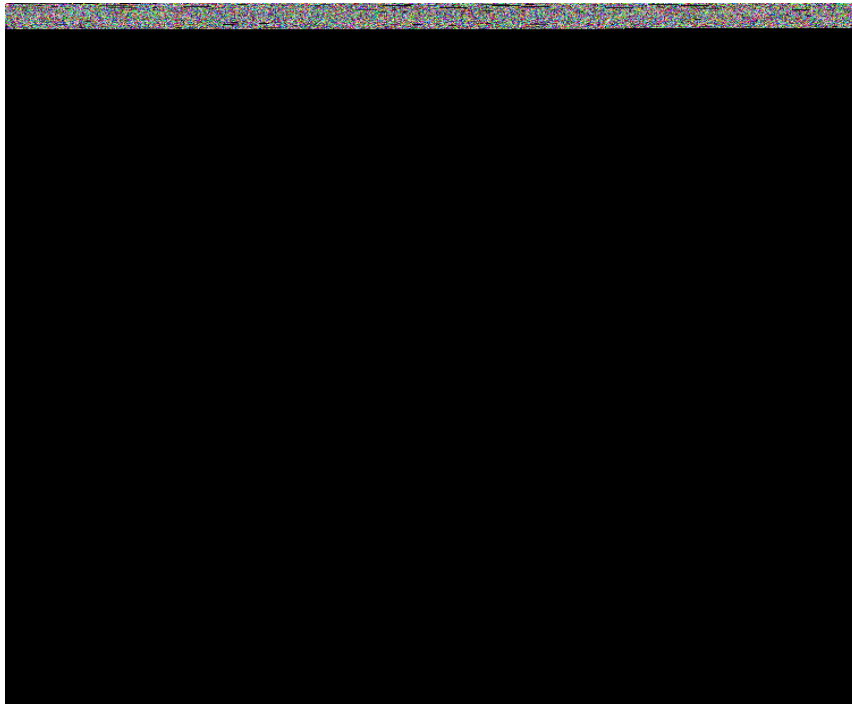


Measurement of properties is independent of peak loads measured

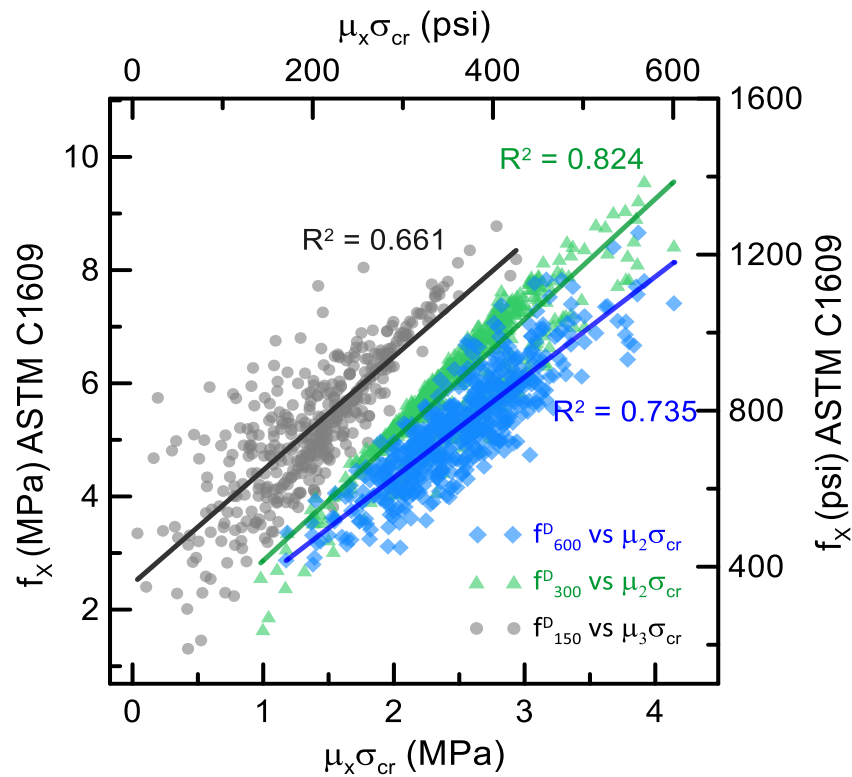




# Tensile Properties, 800+ samples



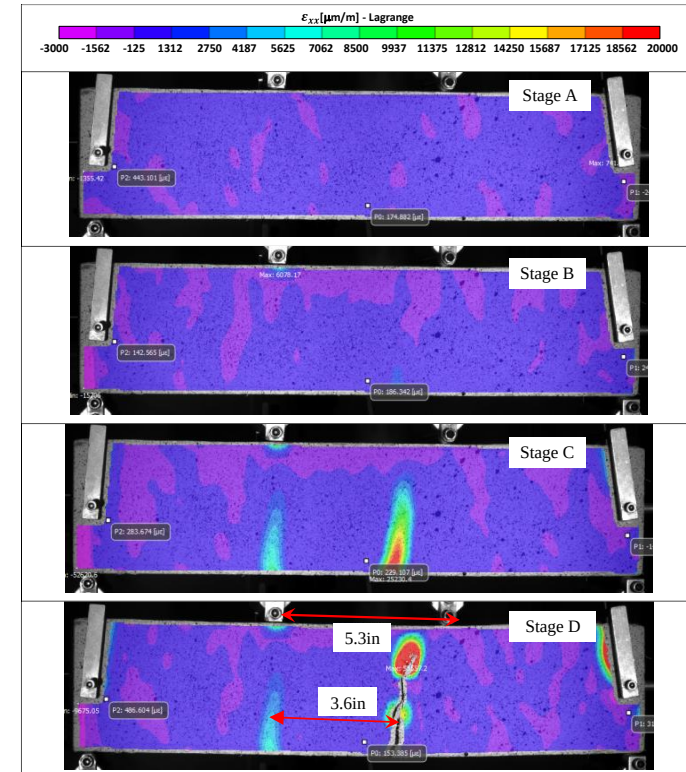
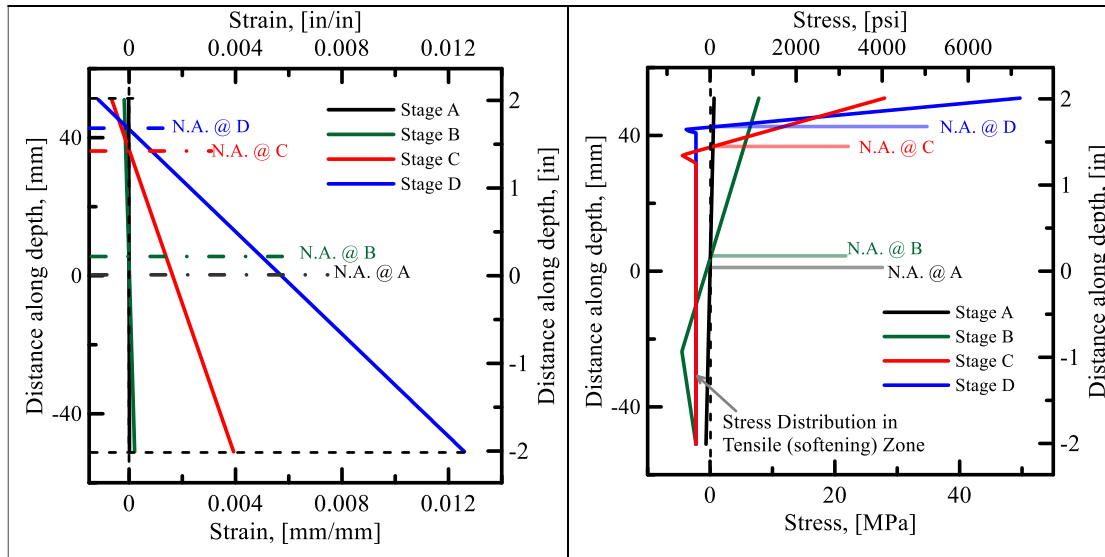
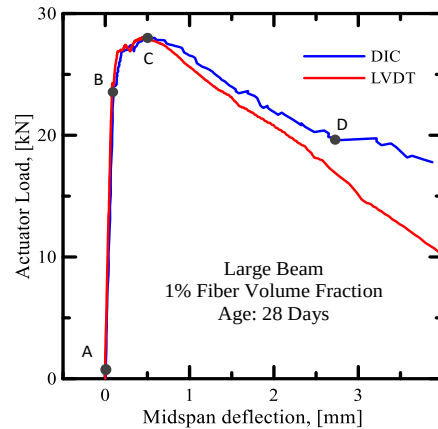
# Correlation of residual Strength with the tensile model



# Applications of the UHPC Model

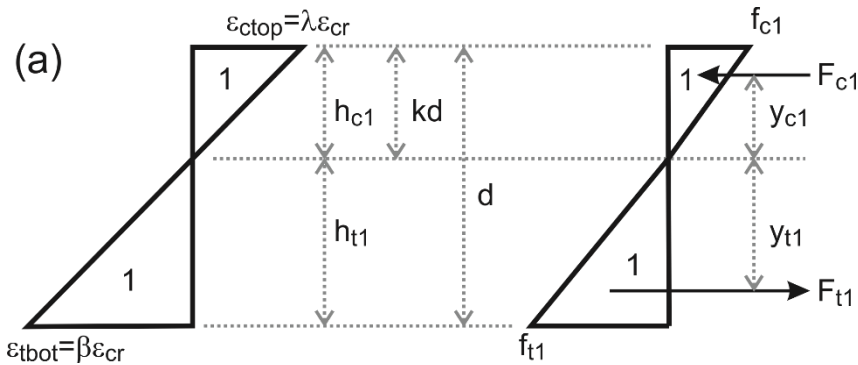
- Validation of the DIC Measurements in Flexural Crack Growth

| Stage | Characteristic                          |
|-------|-----------------------------------------|
| A     | Initiation of the deformation           |
| B     | Initiation of the non-linear response   |
| C     | Response to the peak load               |
| D     | End of DIC data analysis (beam failure) |

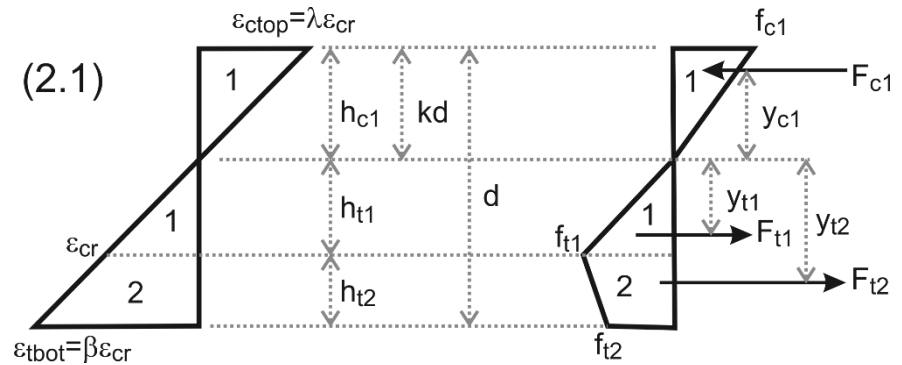


# Stress and Strain Distribution

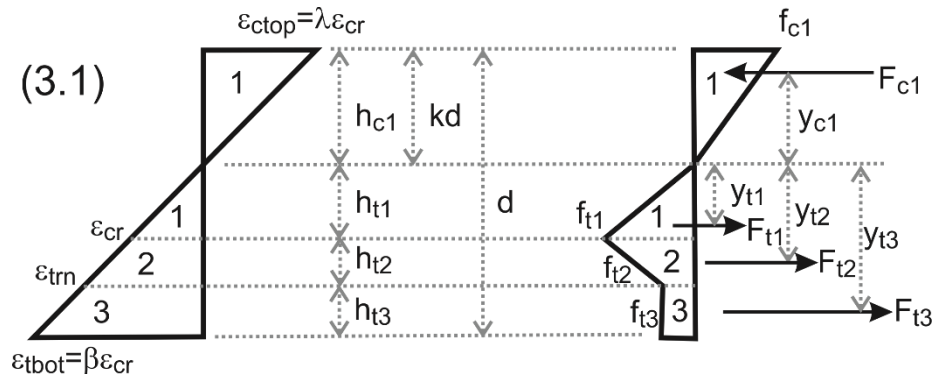
$$0 < \beta \leq 1 \text{ and } \lambda \leq \omega$$



$$1 < \beta \leq \alpha \text{ and } \lambda \leq \omega$$



$$\alpha < \beta \leq \beta_{tu} \text{ and } \lambda \leq \omega$$

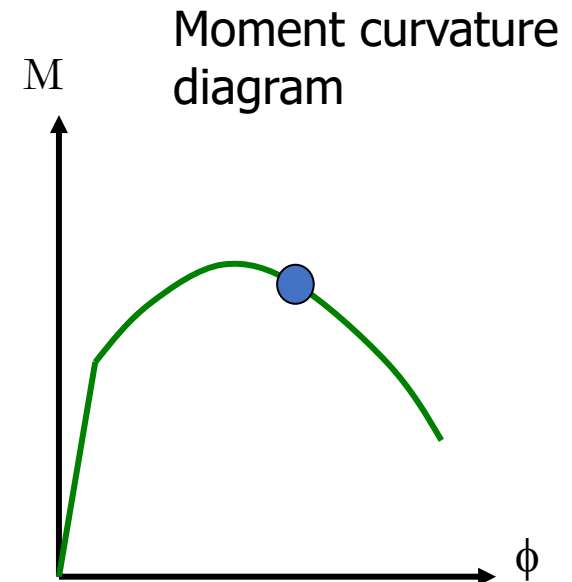
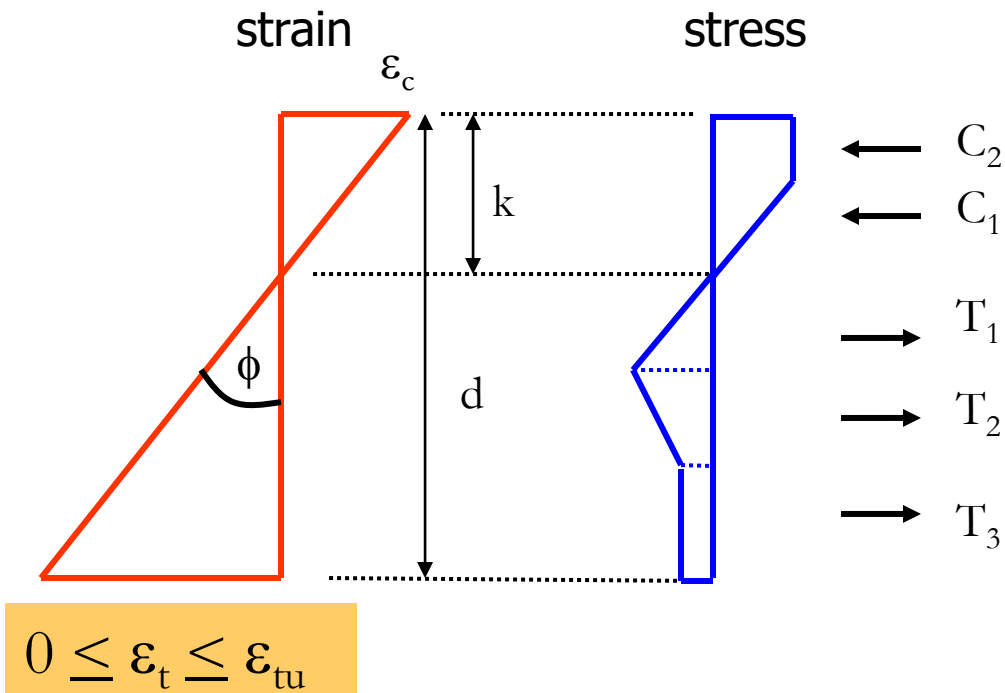


# Moment-Curvature Diagram

- Incrementally impose  $0 \leq \varepsilon_t \leq \varepsilon_{tu}$
- Strain Distribution
- Stress Distribution
- $\Sigma F = 0$ , determine  $k$  (Neutral axis)
- $M = \Sigma C_i y_{ci} + \Sigma T_i y_{ti}$  and  $\phi = \varepsilon_c / kd$
- Normalization  $M' = M/M_0$  and  $\phi' = \phi / \phi_{cr}$

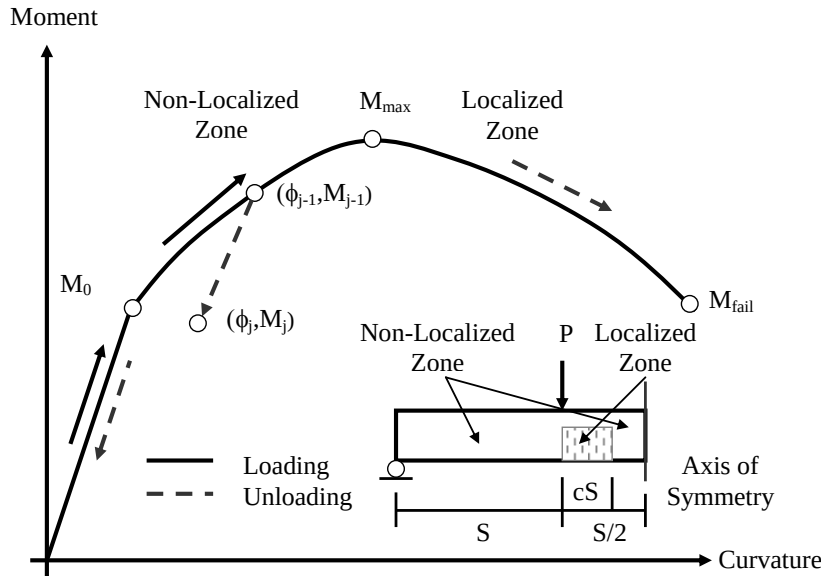
$$F_{c1} = b \int_0^{kd} f_{c1}(y) dy$$

$$y_{c1} = \frac{b}{F_{c1}} \int_0^{kd} f_{c1}(y) y dy$$

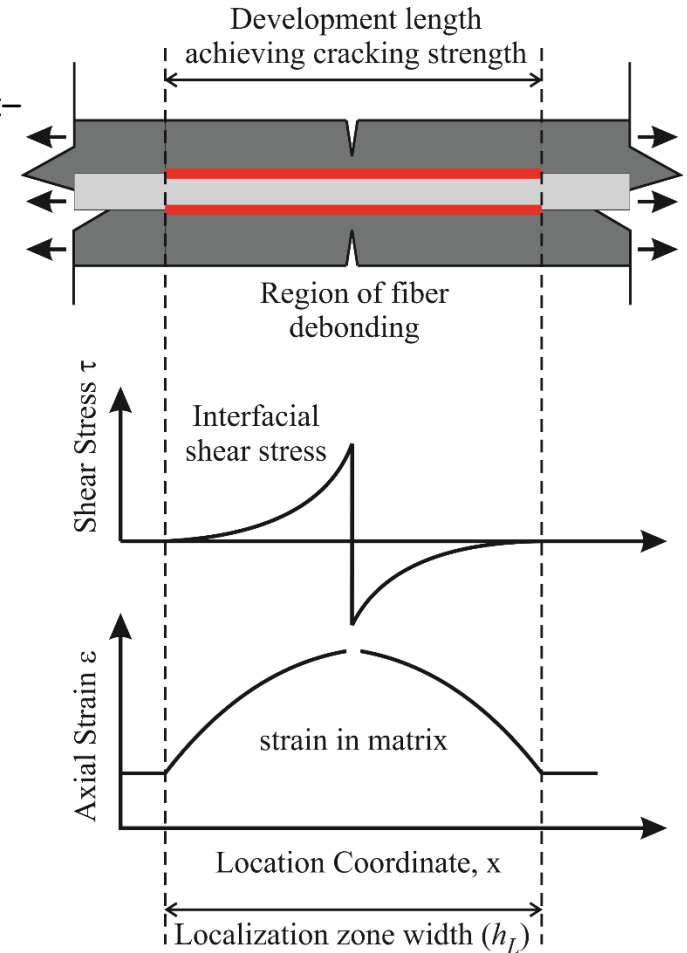


# Localization Rules

- Flexural specimen is loaded beyond the peak strength;
- Distinct zones develop as the deformation localizes in the cracking region an average response over the crack spacing region ;
- Results are used as a smeared crack in conjunction with the moment–curvature diagram
- Measurement from DIC

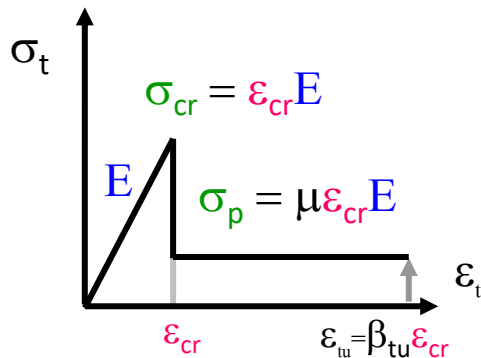


$$\phi_j = \phi_{j-1} - \xi \frac{(M_{j-1} - M_j)}{EI}$$

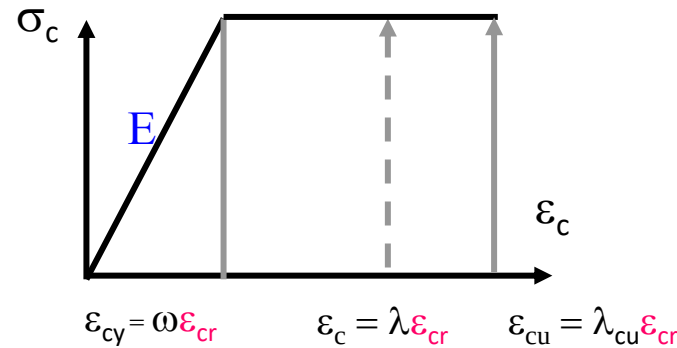


# Material Model for strain softening for serviceability limits

- Two material parameters ( $\epsilon_{cr}$ ,  $E$ ) and four normalized parameters ( $\omega$ ,  $\mu$ ,  $\lambda_{cu}$ ,  $\beta_{tu}$ ), independent variable  $\lambda$ .



Tension model



Compression model

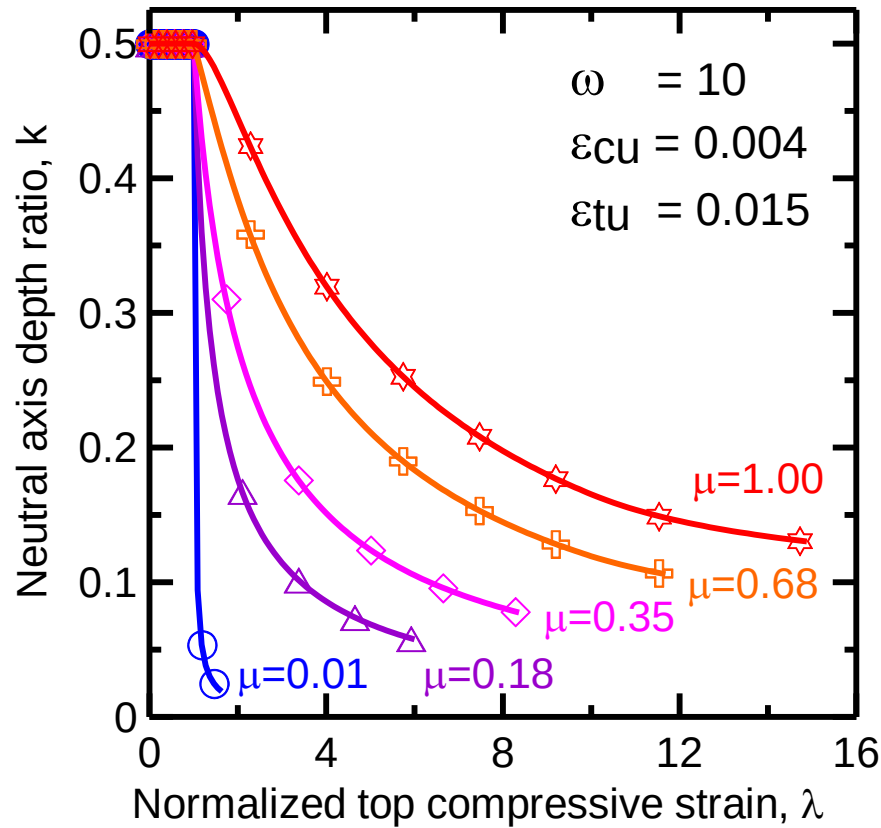
Limit State Design

$$M_{ult} = \left[ 3 \frac{\mu \omega}{\mu + \omega} \right] \frac{1}{6} b d^2 E \epsilon_{cr}$$

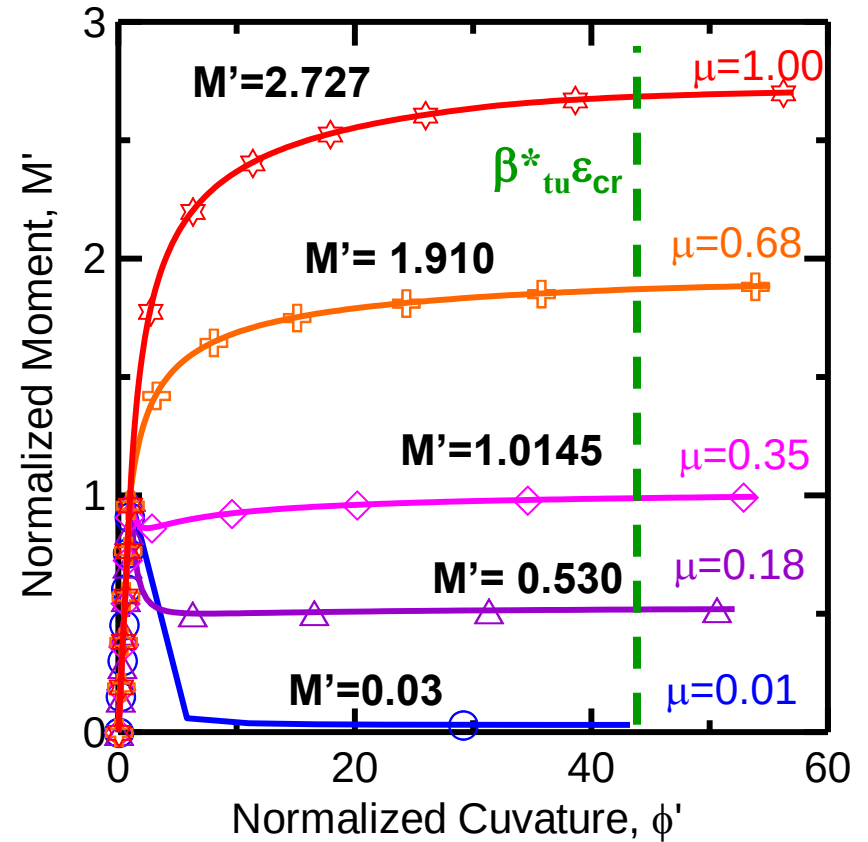
$$M_{cr} = \frac{1}{6} b d^2 E \epsilon_{cr}$$

$$\phi_{cr} = \frac{2 \epsilon_{cr}}{d}$$

# Effect of Residual Strength, $\mu$



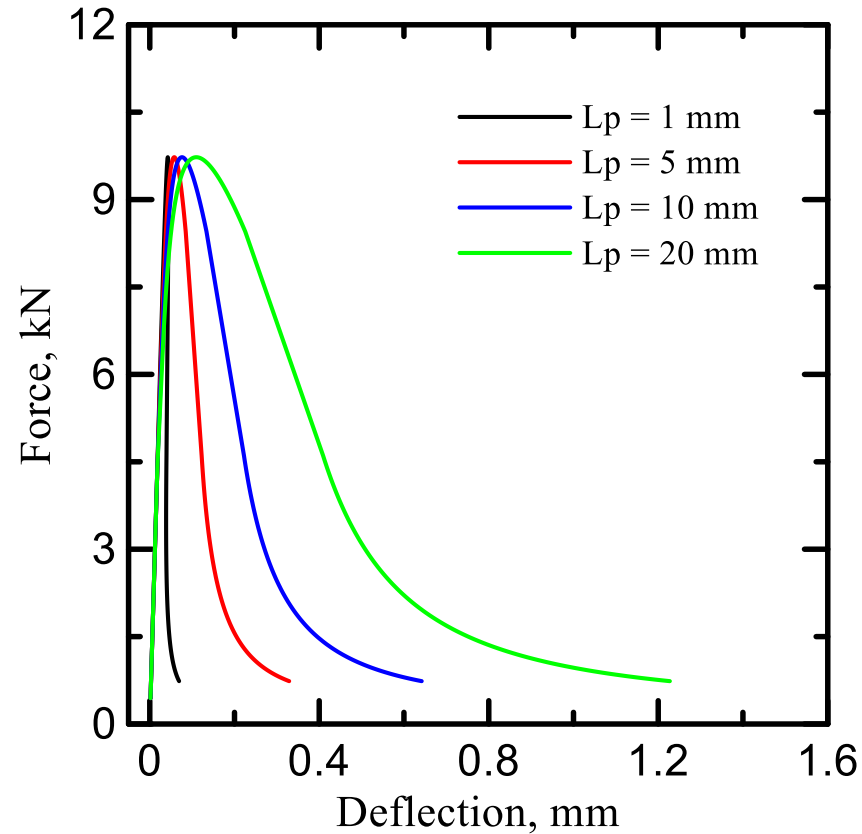
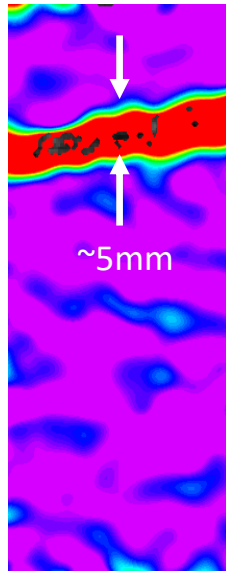
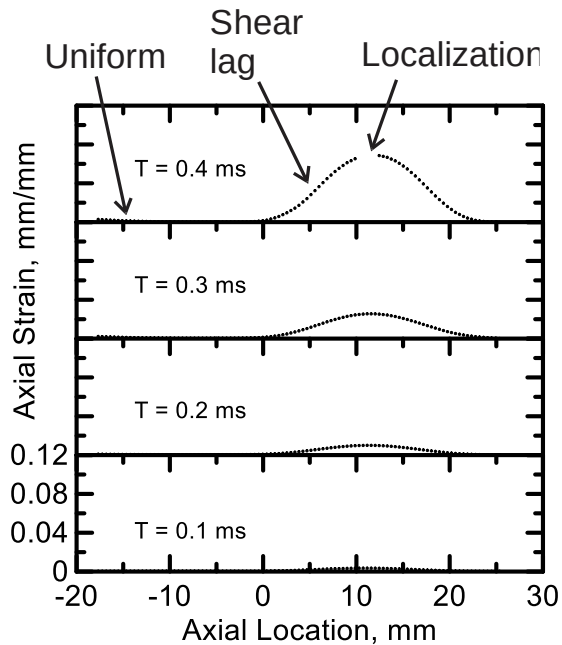
$$M'(\phi \rightarrow \infty) = 3 \frac{\mu\omega}{\mu + \omega}$$



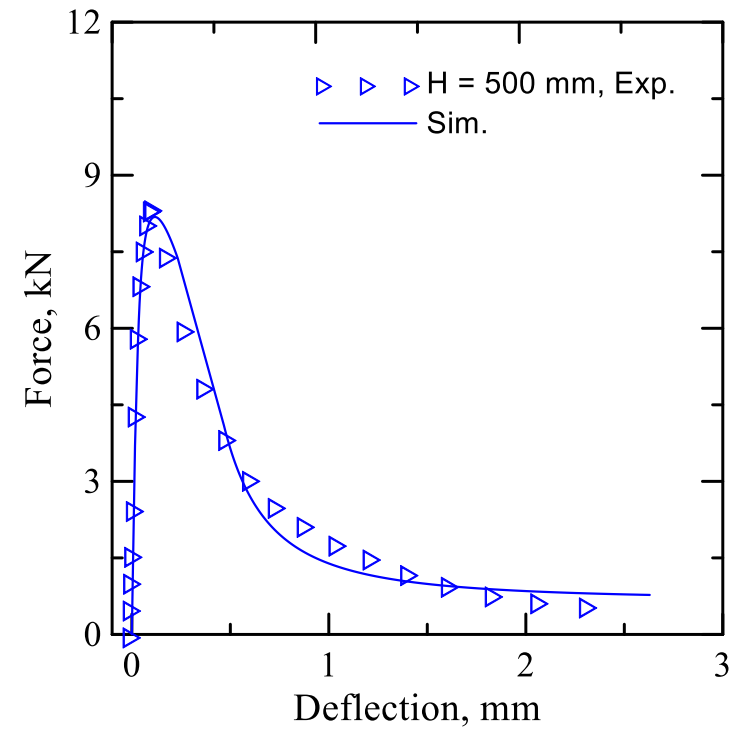
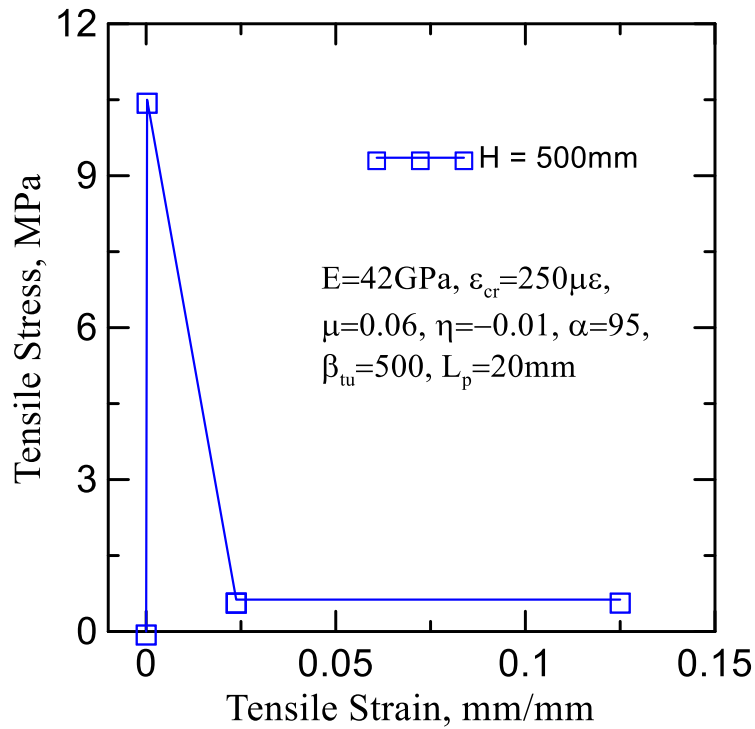
$$M_{cr} = \frac{1}{6} b d^2 E \varepsilon_{cr} \quad \phi_{cr} = \frac{2 \varepsilon_{cr}}{d}$$

# Effect of Localization Zone Size, $L_p$

- Affects the general softening behavior in the post peak zone
- The simulated residual load capacity is not sensitive
- Model parameters:
- $b=40\text{mm}$ ,  $h=40\text{mm}$ ,  $L=125\text{mm}$
- $E=45\text{GPa}$ ,  $\varepsilon_{cr}=280\mu\varepsilon$ ,  $\alpha=90$ ,  $\eta=-0.011$ ,  $\mu=0.021$ ,  $\beta_{tu}=300$ ,  $\gamma=0.95$ ,  $\omega=8.33$ ,  $\lambda_{cu}=27$



# Simulation of Experiment Results

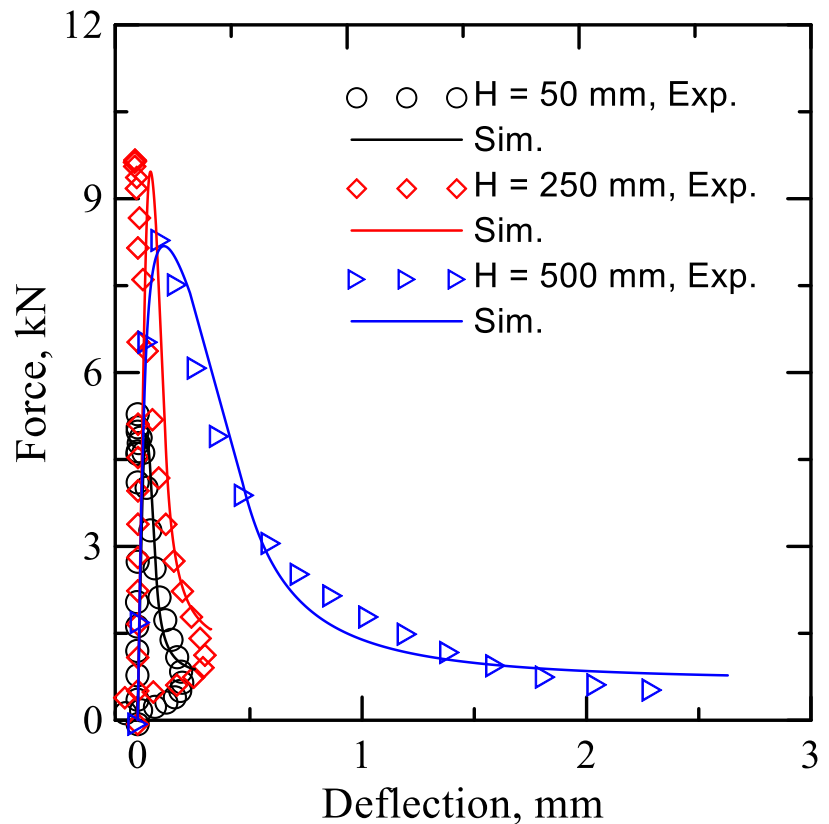
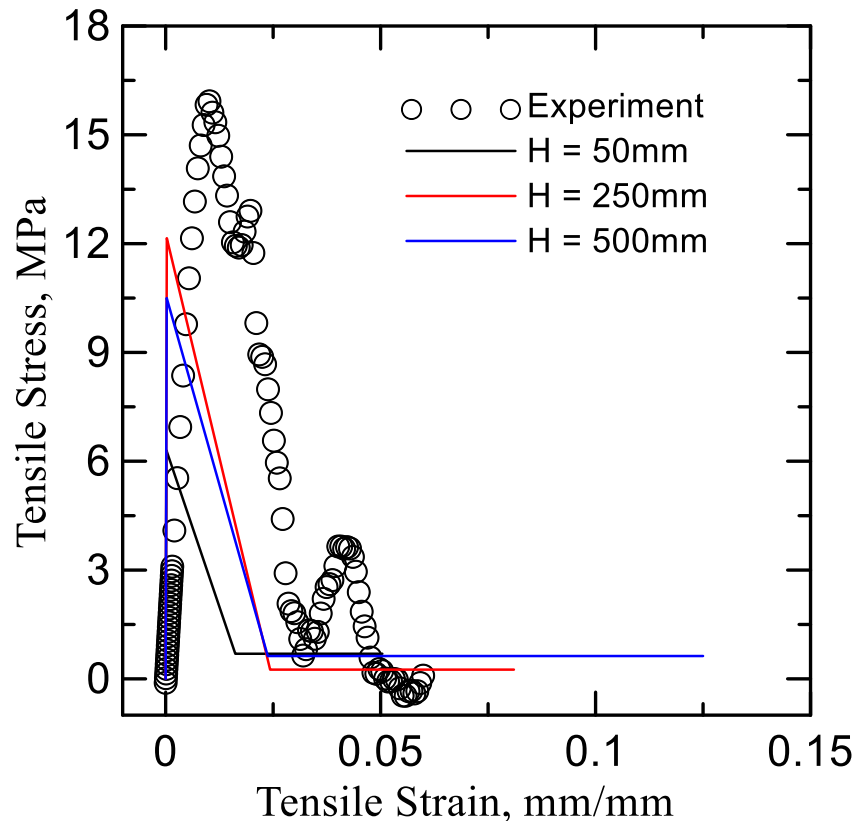


| Drop height (mm) | E (Gpa) | $\varepsilon_{cr}$ ( $\mu\varepsilon$ ) | $\mu$ | $\eta$ | $\alpha$ | $\beta_{tu}$ | $L_p$ (mm) |
|------------------|---------|-----------------------------------------|-------|--------|----------|--------------|------------|
| 50               | 35      | 180                                     | 0.11  | -0.010 | 90       | 280          | 5          |
| 250              | 45      | 270                                     | 0.02  | -0.011 | 90       | 300          | 5          |
| 500              | 42      | 250                                     | 0.06  | -0.010 | 95       | 500          | 20         |

- Other model parameters:
- $\gamma=0.95, \omega=8.33, \lambda cu=27$

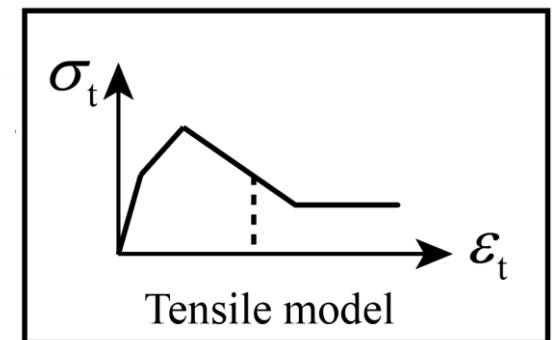
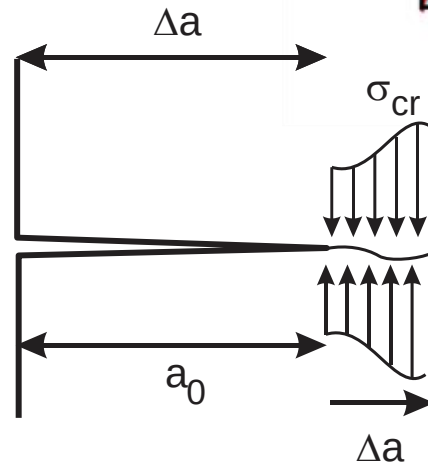
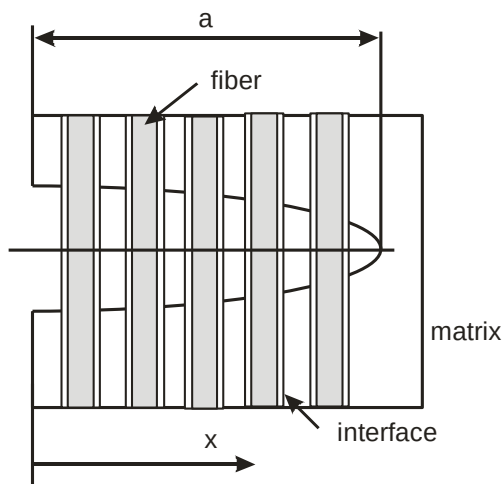
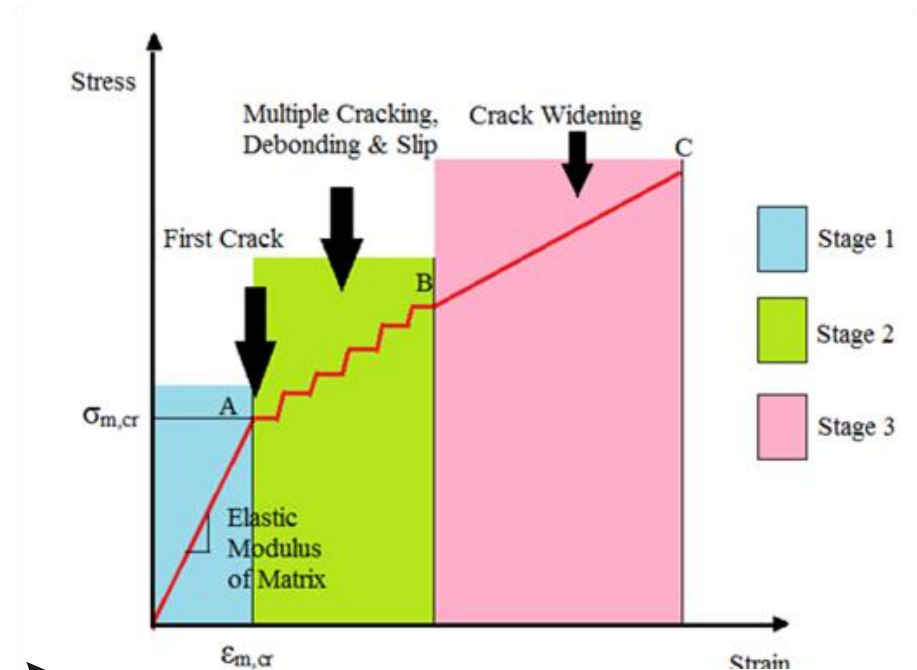
# Correlation of Tensile and Flexural

- Young's Modulus ( $E$ ) and cracking strain ( $\varepsilon_{cr}$ ) increases
- Increasing ultimate tensile strain ( $\beta_{tu}$ ) indicate higher ductility
- Larger localization zone  $L_p$  implies higher level of localized damage, correlating to the failure mode

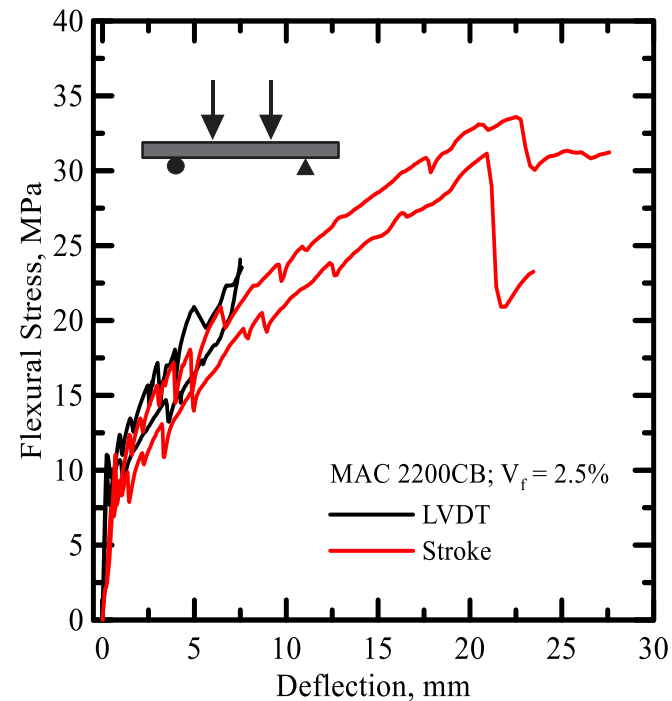
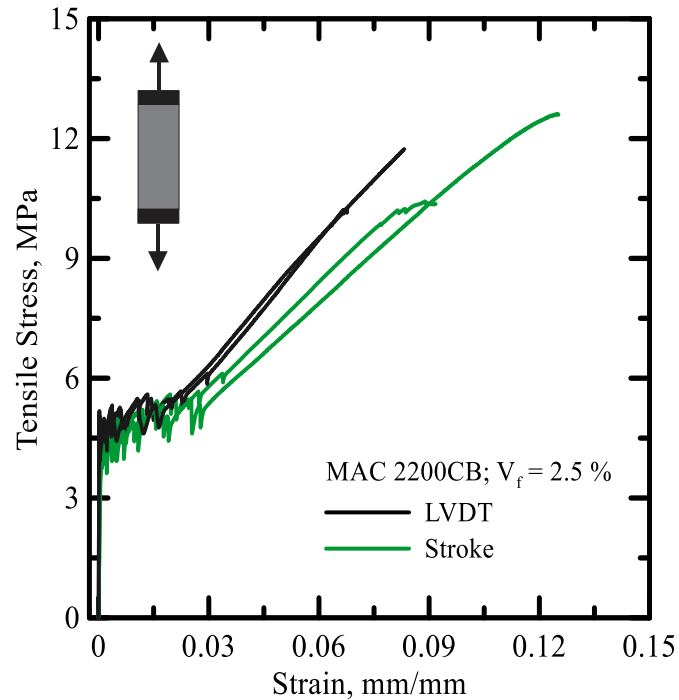


# Typical Stress Strain Response

- **Stage 1** – Linear Elastic region
  - Elastic Modulus
  - First Cracking Stress and strain
- **Stage 2** – Multiple Cracking
  - After first crack propagates through the width.
  - Overall stiffness
  - Fibers bridge cracks
  - Fibers debond
- **Stage 3** – Crack Widening and localization
  - Interface is fully debonded
  - Fiber slipping causes crack widening

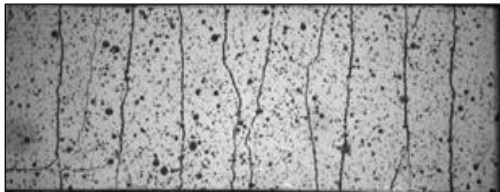
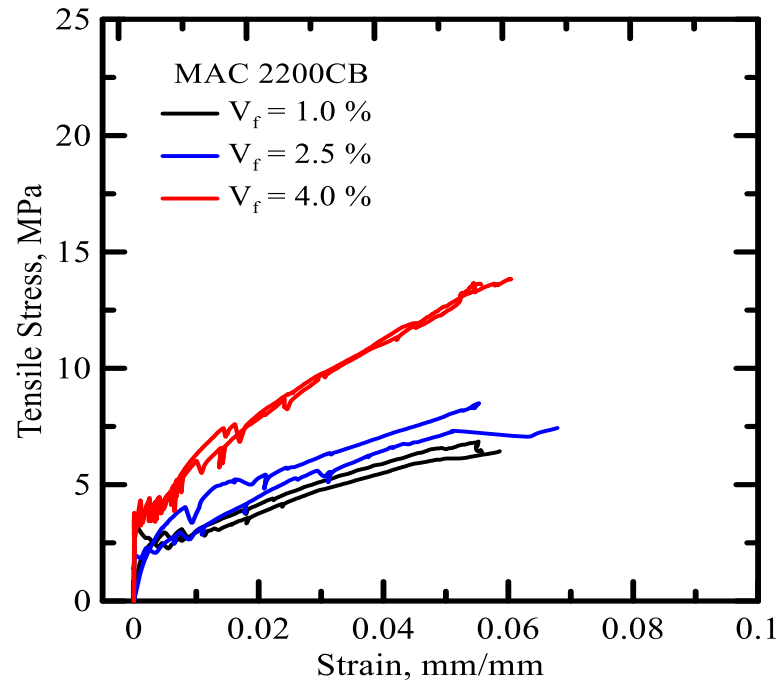


# LVDT versus Stroke Strain- Tension vs. Flexural Response

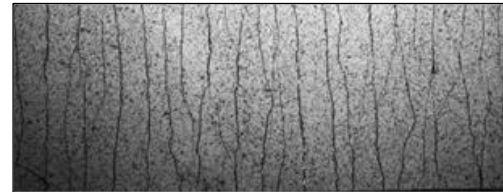
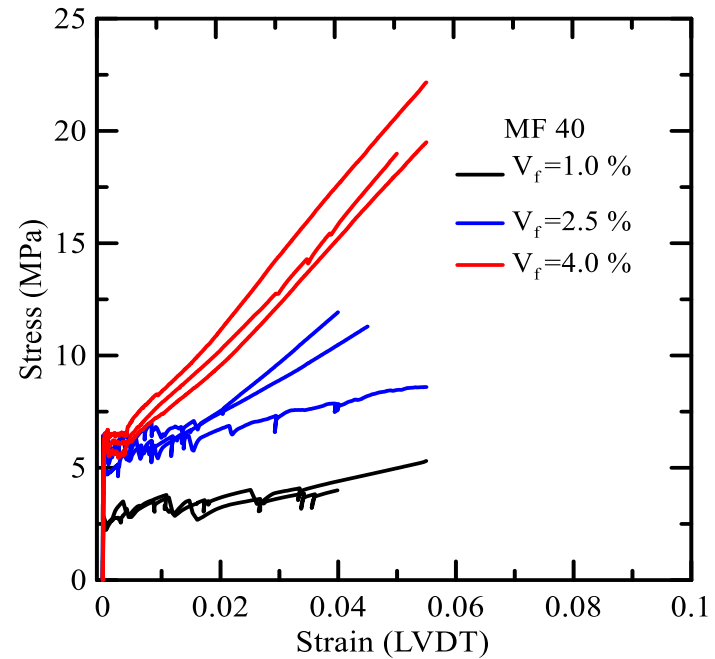


- LVDT - local deformation
- Stiffness, Elastic and Tangent modulus
- Stroke response – Toughness
- Fundamental difference between eth tensile strength and flexural strength

# Effect of Fiber Volume Fraction Control Mix

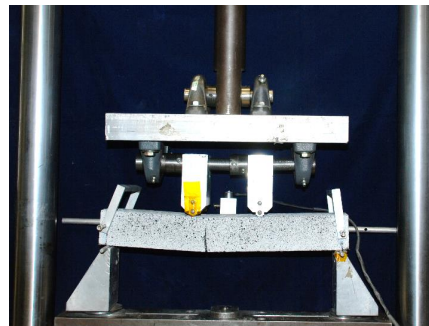
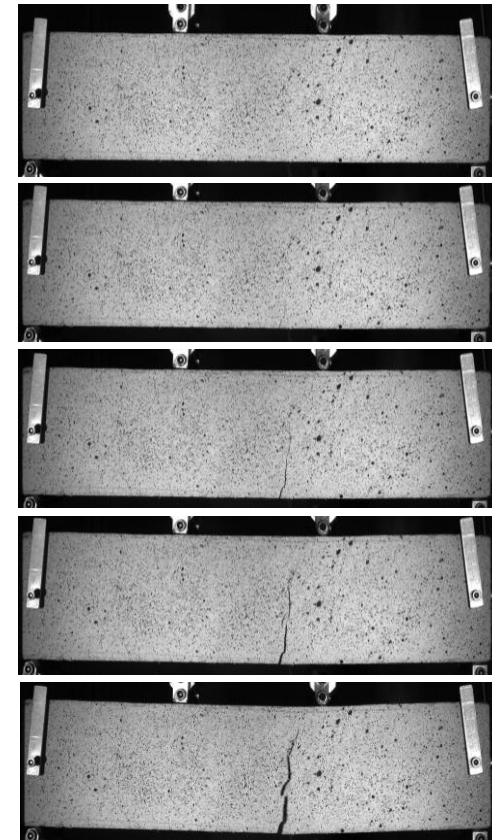
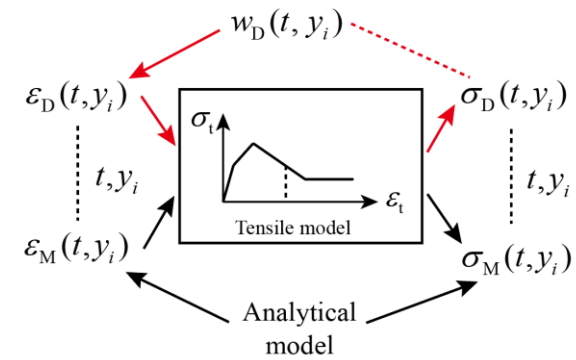
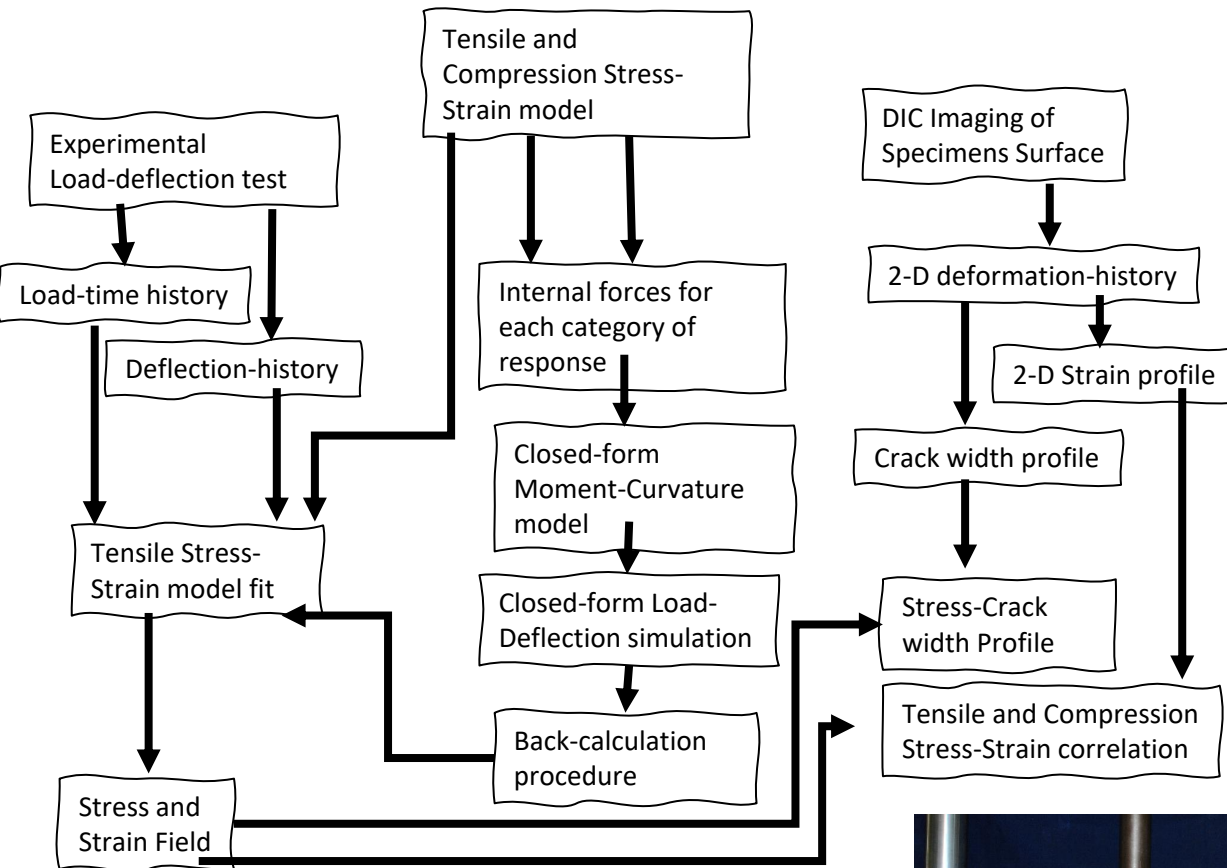


MAC 4% (#7)

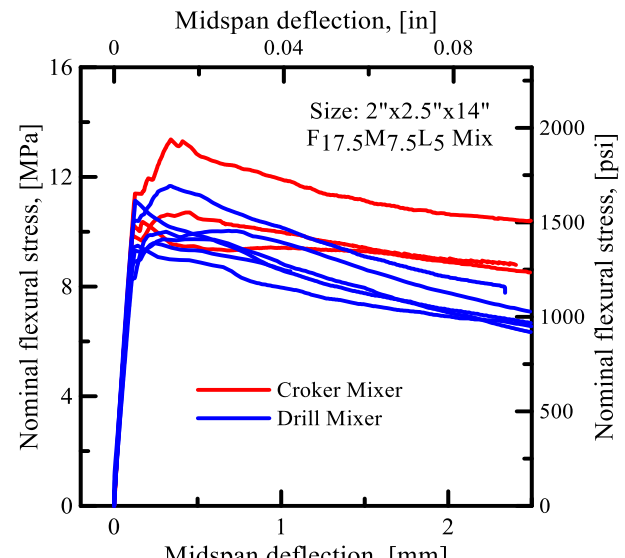
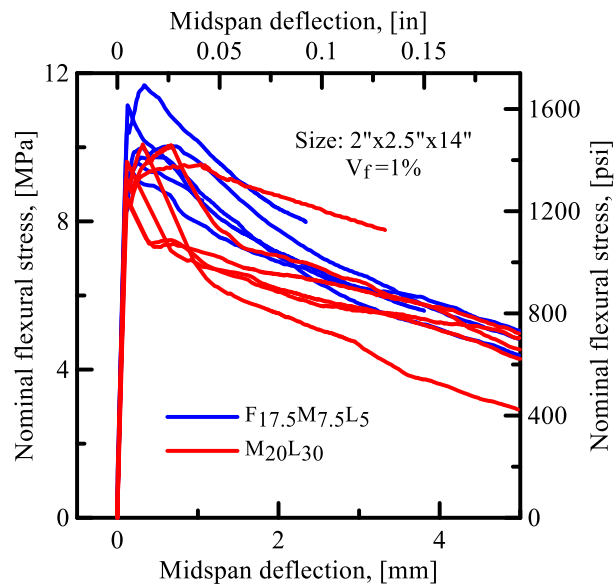
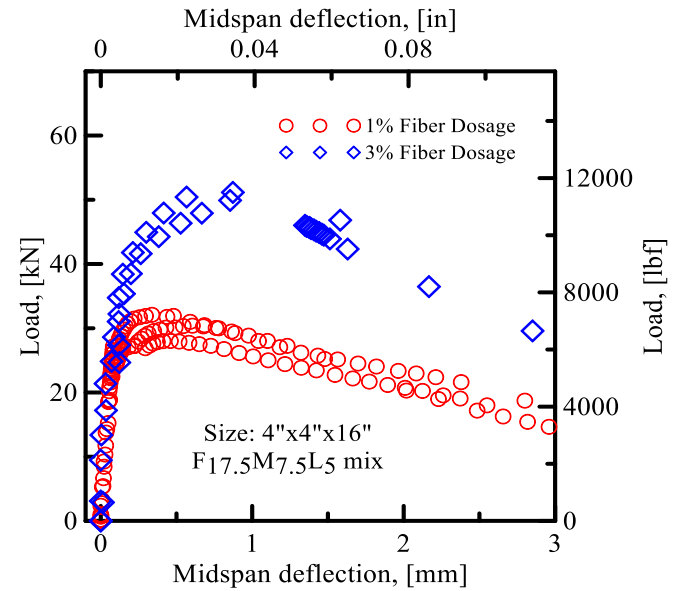
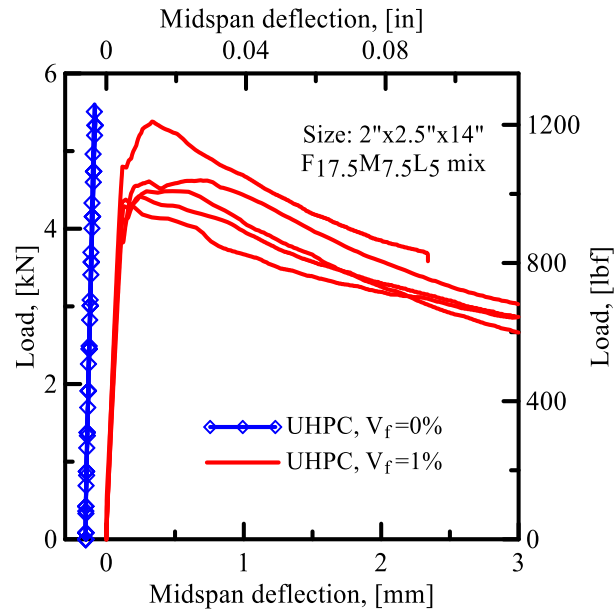


MF40 4% (#3)

Correlation between tension and flexure tests is very useful in confirmation of the results

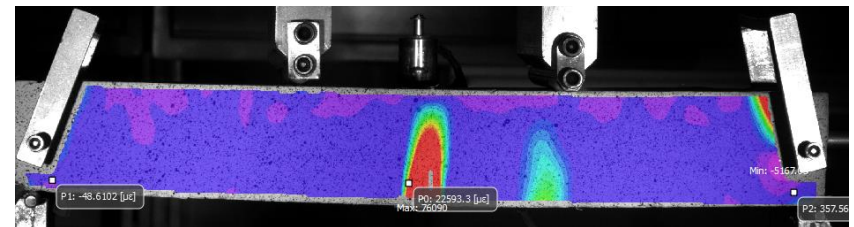
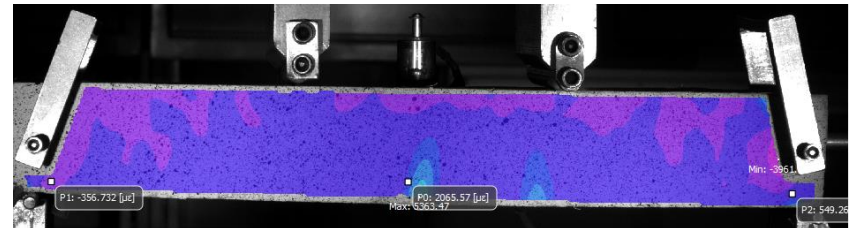
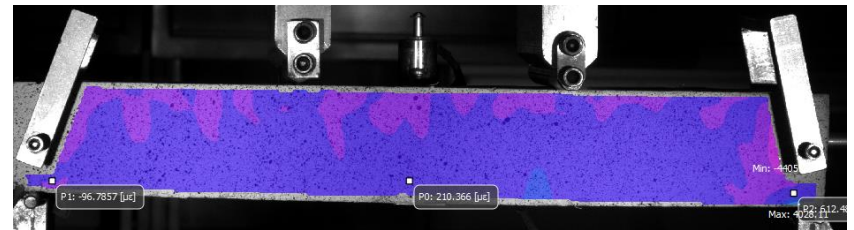
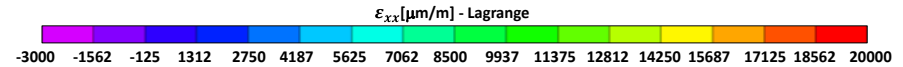
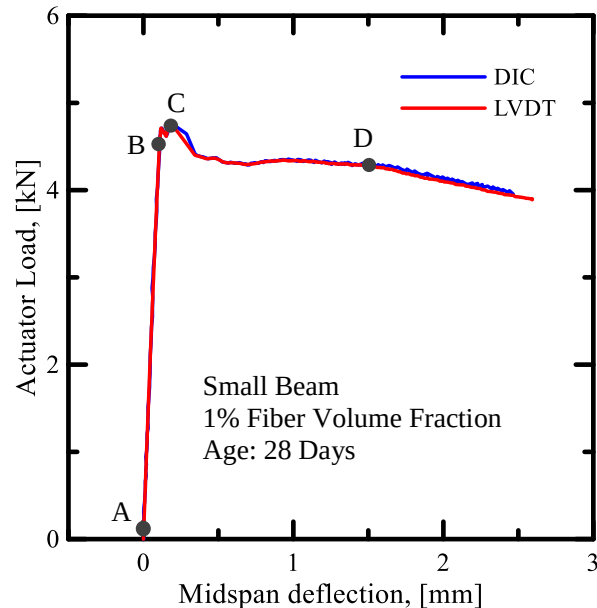


# UHPC in Flexure



Use the DIC crack width and length to correlate the stress crack width models used in the analytical tests

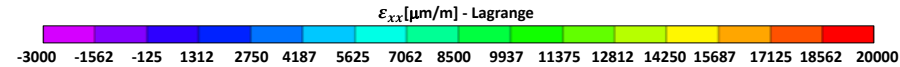
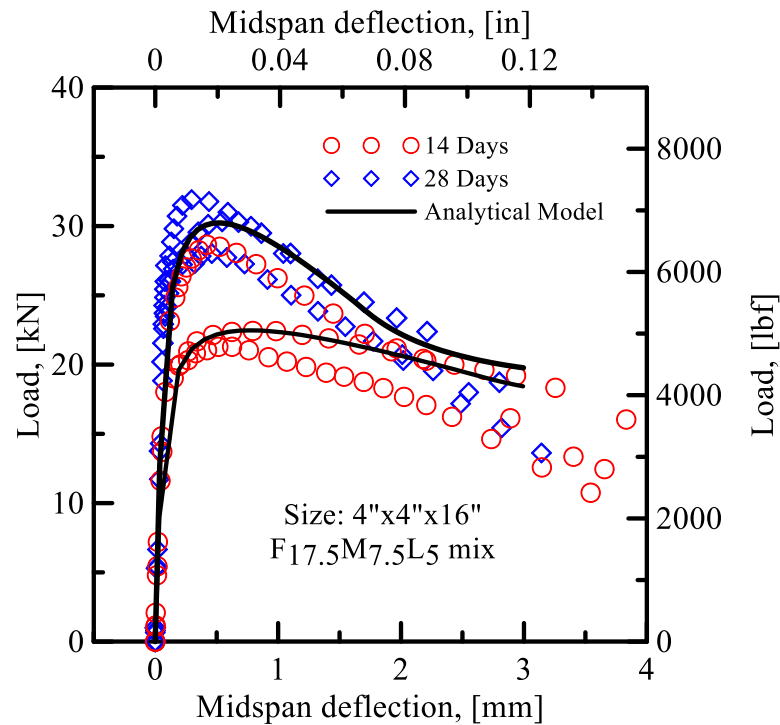
UHPC with 1% fibers, small specimens



Stage D

4.7in  
↔

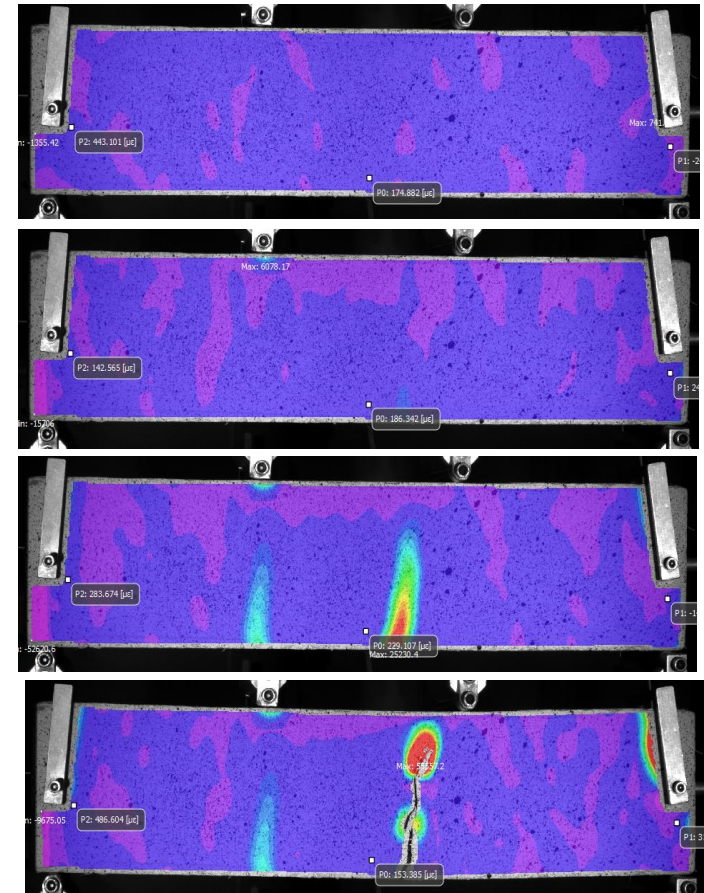
# UHPC large Specimens, more opportunity for the bridging zone to develop



Stage B

Stage C

Stage D



5.3in

# Conclusions and recommendations -I

- The initiation and propagation of a crack in UHPC are two fundamentally different events
- The test method in tension is the most fundamental property and provides significant insight, but there are many parameters that affect the result.
- Boundary conditions such as fixed or rotation allowed affect the results.
- Multiple cracking is a structural response but it affects the strain capacity and post crack stress measurements
- Using notched specimen takes away from the mechanisms of load redistribution and underestimates the ductility of the sample
- Using a dogbone specimen and prismatic specimen also affect test results
- If you use two LVDTs to measure test elongation, it is best to collect individual data signals and compare them for measurement of sample bending during the test
- Cracks form in a uniaxial sample may not open at the same rate. The definition of average crack width is quite arbitrary
- The spacing of two cracks is an important property related to characteristic length of the sample and deserves a lot more attention in our discussions.
- The crack spacing is an indirect measure of interface load transfer between the fiber and matrix.
- Size of the specimen should allow for at least three to four parallel cracks in a prismatic specimen to be used as a valid representation of distributed cracking
- The smaller the crack spacing, the more efficient the load transfer between the fibers and matrix.

## Conclusions and recommendations-II

- Size of the specimen is an important variable, the larger the depth of flexural and length of tension sample, the easier to capture the real stress-crack width relationship and hardening response
- The larger the specimen, the more accurate the results. This has to do with the ability of the fracture process zone to form into a unique size applicable to steady state cracking.
- Small flexural samples underestimate the ductility of the sample significantly
- Notched tensile test is easy to conduct, but the results take away from selection of proper design methodology (we underestimate the sample response by forcing the crack in one location)
- The correlation between the tension and flexural tests provides significant insight into the representative volume element size and the nature of stress distribution.
- Backcalculation of flexural response of load-deflection response to obtain apparent stress-crack width response is a useful tool in the measurement of tensile response applicable to flexural design of UHPC.
- By the time we capture the tension hardening response, measurement of the softening becomes unimportant

# Conclusions and recommendations-III

- Strain softening and hardening and deflection softening and hardening are important characteristics of response in UHPC.
- A strain softening sample can exhibit multiple cracking and deflection hardening in flexure. This means that we can get a significantly ductile flexural sample such as a panel loaded in a biaxial mode (slab restrained around the edges)
- Such structural tests as mentioned above are extremely insightful in calibrating design applications, especially when the structure is designed and forced to behave as an indeterminate continuous slab.
- A useful engineering response in terms of an apparent tensile-stress-strain response can be simulated using a trilinear response.
- A quadlinear tensile-stress-strain response is sufficient for plain UHPC.
- As the UHPC is used in combination with rebars in a Hybrid manner, a trilinear response is sufficient since the rebars provide significant load carrying capacity.

# Some Publications

- Yao, Y., Bakhshi, M., Nasri, V., Mobasher, B., “Interaction diagrams for design of hybrid fiber-reinforced tunnel segments,” February 2018, Materials and Structures 51(1) <https://DOI.org/10.1617/s11527-018-1159-2>
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